Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis Note — Run 1

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Abstract

We present a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion (CCπ^+) differential cross sections on argon-40 using 3.283×10^{20} protons on target (POT) of MicroBooNE data collected in the NuMI Forward Horn Current (FHC) beam during Run 1. Events are selected requiring a reconstructed muon candidate track and exactly one contained pion candidate track emerging from a fiducial neutrino vertex, with zero reconstructed electromagnetic showers and a muon–pion opening angle less than 2.6 rad. In the measured phase space the selection achieves a signal purity of 54.9% and an efficiency of 15.6% (full Run-1 statistics); the lower-threshold reference phase space used for selection development gives 57.2% and 12.1–12.6% (Table 10). Differential cross sections are extracted via Wiener-SVD unfolding in five kinematic variables: muon momentum p_μ, pion momentum p_π, muon polar angle $\cos\theta_\mu$, pion polar angle $\cos\theta_\pi$, and the muon–pion opening angle $\theta_{\mu\pi}$. The observed data yield is approximately $0.75\times$ the GENIE v3 MC prediction, consistent with the known over-prediction of CCπ^+ production in current neutrino interaction generators. Using the confirmed integrated $\nu_\mu + \bar{\nu}_\mu$ flux of 6.81159×10^{-10} cm^{-2} POT^{-1} (the flux-file author's GEANT4 value integrated above 60 MeV, superseding the 2.3725×10^{-9} read from the flux histograms, which was a factor 3.48 too large), the Wiener-SVD integrated cross section from the angular observables is $\sigma_{\text{int}} \approx 1.12 \times 10^{-38}$ cm^2/Ar. The extraction is validated by a same-sample self-closure that recovers the truth to within 6% for every observable and by exact algorithmic equivalence with the official XSecAnalyzer framework unfolder (Sec. B.1), confirming that the fake-data offsets are POT bookkeeping and genuine model differences, not unfolding bias. All differential results, figures, and systematics use the XSecAnalyzer framework extraction on the resolution-optimised binning of Sec. 13; this is the single consolidated analysis note.

Contents

1 Introduction

Charged-current single charged-pion (CCπ^+) production,

$$\nu_\mu(\bar{\nu}_\mu) + \mathrm{Ar} \longrightarrow \mu^\mp + \pi^\pm + X, \tag{1}$$

is among the most frequent neutrino interaction channels at energies of 1–5 GeV and constitutes a significant background for oscillation analyses at the same energy scale, most notably for ν_e appearance measurements at experiments such as NOvA [1] and T2K [2], as well as the future DUNE [3]. Despite its importance, the cross section on nuclear targets remains poorly constrained: major neutrino interaction generators (GENIE v3, NuWro, NEUT) disagree by 20–30% in absolute rate and show significant shape differences for individual kinematic variables.

MicroBooNE is a single-phase liquid argon time-projection chamber (LArTPC) at Fermilab that operated from 2015–2021, providing the world's highest-statistics ν_μ-on-argon dataset below 5 GeV during its NuMI running period. This note presents a simultaneous measurement of five differential cross sections in the CCπ^+ channel using Run 1 NuMI FHC data, employing Pandora pattern recognition [4] and Wiener-SVD unfolding [5].

2 Beam and Neutrino Flux

2.1 NuMI beam and flux composition

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector striking a graphite target. Charged secondary hadrons are focused by two magnetic horns whose polarity selects the sign of the focused mesons. In Forward Horn Current (FHC) mode the horns focus positive pions and kaons, giving a beam with predominantly ν_μ content; in Reverse Horn Current (RHC) mode the polarity is reversed to focus negatives, giving a predominantly $\bar{\nu}_\mu$ beam. MicroBooNE is located approximately 110 mrad off-axis from the NuMI beam axis, reducing the peak neutrino energy to the 0.5–2.5 GeV range and narrowing the energy spectrum. Because MicroBooNE sits off-axis and at comparatively low energy, the horn sign-selection is imperfect and both modes carry a sizeable wrong-sign component — most notably RHC, whose wrong-sign ν_μ fraction is large.

The flux composition at the MicroBooNE site is taken from the new-G4 central-value flux histograms (`flux_newg4.root`), integrated over $E_\nu > 60$ MeV to exclude an unphysical sub-threshold pile-up near $E_\nu = 0$ (removed by the flux group's `RemoveFluxPeak` step; below the reaction threshold and irrelevant to the measurement). The muon-flavour split of these central-value histograms matches the author's authoritative integrated fluxes exactly (Sec. 2.3):

Table 1: NuMI flux composition at the MicroBooNE site by neutrino type, for the FHC and RHC horn modes (new-G4 central-value flux `flux_newg4.root`, integrated over $E_\nu > 0.1$ GeV). FHC is ν_μ-dominant; RHC is $\bar{\nu}_\mu$-dominant with a large wrong-sign ν_μ fraction. The combined row weights each mode by its data exposure (FHC 8.857, RHC 11.082×10^{20} POT; Table 2), giving a near-balanced $\nu_\mu/\bar{\nu}_\mu$ mix. The signal is charge-blind ($|\nu_{\mathrm{pdg}}| = 14$), so both ν_μ and $\bar{\nu}_\mu$ contribute in every mode.

Horn mode	ν_μ	$\bar{\nu}_\mu$	ν_e	$\bar{\nu}_e$
FHC	63.4%	34.0%	1.7%	0.9%
RHC	44.2%	53.3%	1.4%	1.2%
Combined	53.0%	44.4%	1.5%	1.1%

Flux corrections from the Package to Predict the FluX (PPFX) [6] are applied per event via the central-value weight `ppfx_cv` stored in the analysis ntuples. (The absolute flux normalisation used for the cross-section in Sec. 2.3 is still taken from the previous flux histograms; moving that normalisation onto the new-G4 flux is a separate, tracked update.)

2.2 Data exposure

The analysis is *blind*: the real beam-on data are not used. Each configuration is extracted on Poisson-thrown central-value fake data scaled to the corresponding ν_e-matched data POT (Table 2). The FHC measurement uses the full FHC exposure (Runs 1, 2, 4 and 5); RHC uses Runs 1, 2, 3, 4a, 4b and 4c; the combined measurement uses all fourteen overlay files, each horn mode scaled to its own data exposure.

Table 2: Run-period exposure accounting (final, full FHC+RHC). "MC POT" is each period's native simulated exposure; "Data POT" is the ν_e-matched exposure to which the (blind) fake data is scaled. The runs of a given horn mode are *pooled* into a single sample: every file of the mode is scaled by the *same* factor, Scale = (mode Data POT)/(mode *total* MC POT), rather than each file being scaled by its own POT (which would over-count). A run with larger POT already contributes proportionally more events, so this single-factor pooling weights each run by its statistics and yields the correct rate. The combined measurement applies the FHC factor (0.0924) to the FHC files and the RHC factor (0.0717) to the RHC files. Run 5 uses the reweighted-PPFX ν_μ overlay.

Run period	MC POT ($\times 10^{20}$)	Data POT ($\times 10^{20}$)	Scale
Run 1 (FHC)	23.282		
Run 2 (FHC)	24.934		
Run 4 (FHC)	28.335		
Run 5 (FHC)	19.300		
FHC total	**95.85**	**8.857**	0.0924
Run 1 (RHC)	8.997		
Run 2 (RHC)	57.865		
Run 3 (RHC)	55.185		
Run 4a (RHC)	7.283		
Run 4b (RHC)	23.798		
Run 4c (RHC)	1.505		
RHC total	**154.63**	**11.082**	0.0717
Combined (FHC+RHC)	**250.49**	**19.939**	(per mode)

The full exposure is a $\approx 2.7\times$ increase over the single-run Run-1 FHC baseline (3.283×10^{20}) in FHC alone, and $\approx 6\times$ combined, substantially reducing the data-statistical term (Tables 19–21). When the analysis is unblinded, the real beam-on POT (and gate counts) replace the ν_e-matched values above.

2.3 Integrated flux

The absolute flux normalisation uses the new-G4 GEANT4 NuMI flux (`flux_newg4.root`), integrated over $E_\nu > 60$ MeV (the flux group's `RemoveFluxPeak` cut, which removes an unphysical ~25–30 MeV artifact spike). The signal is charge-blind ($|\nu_{\mathrm{pdg}}| = 14$, covering both $\nu_\mu \to \mu^- \pi^+$ and $\bar{\nu}_\mu \to \mu^+ \pi^-$), so the normalisation uses the combined $\nu_\mu + \bar{\nu}_\mu$ flux $\Phi = \Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$. Because RHC is $\bar{\nu}_\mu$-dominant with a different absolute flux than FHC, each configuration uses its *own* integrated flux:

Table 3: Authoritative integrated ν_μ and $\bar{\nu}_\mu$ fluxes per POT (new-G4 flux, $E_\nu > 60$ MeV), and the combined $\nu_\mu + \bar{\nu}_\mu$ value used as the per-configuration normalisation. The combined row is the data-POT-weighted average of the two modes (weights 8.857 : 11.082; Table 2). The FHC value 6.81159×10^{-10} is the previously confirmed normalisation; RHC and combined were previously normalised with the FHC flux and are corrected here (raising the RHC cross section by 5.7% and the combined by 3.1%).

Config	Φ_{ν_μ}	$\Phi_{\bar{\nu}_\mu}$	$\Phi_{\nu_\mu+\bar{\nu}_\mu}$	$\bar{\nu}_\mu$ frac.
	($10^{-10}\,\mathrm{cm}^{-2}\,\mathrm{POT}^{-1}$)			
FHC	4.43515	2.37644	6.81159	34.9%
RHC	2.92348	3.52298	6.44646	54.6%
Combined	3.59497	3.01368	6.60865	45.6%

The mean $\bar{\nu}_\mu$ energy is 0.500 GeV (FHC) and 0.417 GeV (RHC), the lower RHC value reflecting the softer wrong-sign-enhanced spectrum. These fluxes are set per configuration in `integrated_numu_flux_in_FV()` (`include/XSecAnalyzer/FiducialVolume.hh`). The correction relative to the superseded flux (`uboone_numi_flux_histograms.root`, whose combined value 2.3725×10^{-9} was $\sim 3.48\times$ too large and made every extracted cross section $\sim 3.5\times$ too small) was validated four independent ways — standalone NuWro and GENIE on the old vs. corrected flux, the flux author's numbers, and a GENIE closure — documented in `generator_predictions/`.

The number of argon target nuclei in the fiducial volume is computed from first principles as

$$N_{\mathrm{Ar}} = \frac{\rho_{\mathrm{LAr}} \cdot V_{\mathrm{FV}} \cdot N_A}{A_{\mathrm{Ar}}} \approx 8.71 \times 10^{28}, \tag{2}$$

and the per-configuration normalisation factor is $N_{\mathrm{norm}} = \Phi_{\mathrm{mode}} \cdot N_{\mathrm{POT,mode}} \cdot N_{\mathrm{Ar}}$, evaluated with each mode's flux and data POT (Table 2).

3 Monte Carlo Samples

Monte Carlo events are generated with GENIE v3 [7] on argon-40 and overlaid with real cosmic-ray data (the "overlay" technique) so that simulated events experience in-situ detector noise and cosmogenic activity. Table 4 summarises the input samples.

Table 4: Input samples (final, full FHC+RHC). "n" is the number of overlay files; "MC POT" is the summed native simulated exposure ($\times 10^{20}$). The blind Poisson-thrown fake data stands in for the (unused) beam-on data; the detector variations use the native Run-4 FHC and RHC samples. The per-file lists and POT/trigger values are in the `configs/file_properties_numi_*` tables.

Sample	Composition	n	MC POT
ν_μ overlay (FHC)	Runs 1, 2, 4 (pandora) + Run 5 (reweighted-PPFX)	4	95.85
ν_μ overlay (RHC)	Runs 1, 2, 3, 4a, 4b, 4c (pandora)	10	154.63
Dirt MC	GENIE NuMI dirt overlay (Run 1)	1	16.74
EXT (beam-off)	cosmic data, per run, horn-mode-independent	7	—
Detector variations	Run-4 FHC + Run-4 RHC, CV + 8 knobs each	18	—
Fake data (blind)	Poisson-thrown central value, per configuration	—	—
Beam-on data	blinded — not used until unblinding	—	—

Event weights. Each MC event is weighted by

$$w = \mathrm{Scale[mode]} \times w_{\mathrm{PPFX}} \times w_{\mathrm{tune}}, \tag{3}$$

where Scale[mode] is the single per-mode POT factor of Table 2 (all files of a horn mode pooled), w_{PPFX} is the PPFX central-value flux correction (`ppfx_cv`) and w_{tune} is the GENIE/GEANT4 reweighting correction (`weightTune`). Events with non-finite, zero, or unphysically large weights ($w > 30$) are assigned unit weight.

EXT scaling. The beam-off cosmics sample is scaled by the ratio of beam-on to beam-off trigger counts:

$$\mathrm{Scale}_{\mathrm{EXT}} = \frac{7\,809\,962}{610\,496 + 3\,211\,097} = 2.044. \tag{4}$$

The uncertainty on this ratio is a leading systematic on the EXT background.

Software trigger. Data and EXT files are pre-filtered to require the software trigger (`swtrig == 1`). MC events failing `swtrig = 0` are removed at the NeutrinoSlice cut stage. In practice this has no measurable effect at the current working point: all MC events passing the topological cut are verified to have `swtrig == 1` (46 810/46 810 events confirmed numerically).

4 Signal Definition

The signal is defined at generator level and is used only to construct the selection efficiency and response matrix. It is *never* applied as a reconstruction cut. The definition is *charge-blind*: both $\nu_\mu \to \mu^-\pi^+$ and $\bar{\nu}_\mu \to \mu^+\pi^-$ are treated as signal on the same footing.

Neutrino flavour
: $|\nu\ \mathrm{PDG}| = 14$ (ν_μ or $\bar{\nu}_\mu$).

Interaction type
: Charged-current (`ccnc == 0`).

True vertex
: Inside the fiducial volume (Section 5).

Primary multiplicity
: Exactly one muon ($|\mathrm{PDG}| = 13$), exactly one charged pion ($|\mathrm{PDG}| = 211$), zero neutral pions ($|\mathrm{PDG}| = 111$), zero kaons. Any number of protons is allowed.

Kinematic thresholds
: $p_\mu > 0.15\,\mathrm{GeV}/c$, $p_\pi > 0.175\,\mathrm{GeV}/c$, and $\theta_{\mu\pi} < 2.6$ rad. These define the *measured phase space*, chosen from the efficiency turn-on curves (Section 7). The momentum efficiency turns on sharply at $\sim$0.15–0.20 GeV/c(set by the 10 cm muon and 20 cm pion track-length cuts); below that it is only $\sim$1–3% with a strongly model-dependent correction. The opening-angle upper bound (2.6 rad) excludes the large-angle region, which is dominated by cosmic and dirt background ($\sim$7% purity in the $\theta_{\mu\pi} > 2.513$ rad bin). No *lower* opening-angle cut is applied: the small-angle region is high-purity ($\sim$58%), so cutting it would only remove signal (it is low-efficiency but not low-purity). The same cut is applied at reco level so the signal and selection acceptances match. The first p_μ/p_π bins and the $\theta_{\mu\pi}$ upper edge start/end at these thresholds.

(Earlier iterations used $p_\mu, p_\pi > 0.1\,\mathrm{GeV}/c$ with the full 0–π angle; that left near-dead efficiency bins at low momentum and at the large-angle extreme. The phase space above raises the efficiency from 12.1% to 15.6% at $\sim$55% purity — see Section 7.)

Pion threshold and the sibling ν_e measurement. The companion MicroBooNE NuMI ν_e CC 1π analysis defines its pion signal by kinetic energy, $\mathrm{KE}_\pi > 40$ MeV, equivalent to $p_\pi > 0.113\,\mathrm{GeV}/c$. Our $p_\pi > 0.175\,\mathrm{GeV}/c$ ($\mathrm{KE}_\pi > 84$ MeV) is therefore a higher threshold. Lowering it toward the ν_e value is possible — the signal definition is a truth-level choice and the unfolding maps reco to truth — but the pion momentum reconstruction (track range under the muon hypothesis, mass-corrected) degrades rapidly below $0.175\,\mathrm{GeV}/c$:

Table 5: Pion momentum reconstruction versus true p_π (correctly reconstructed signal pions, Run 1). Below the $0.175\,\mathrm{GeV}/c$ threshold the bias and resolution blow up and the reconstructed fraction halves.

True p_π [GeV/c]	Bias $\langle (p^{\mathrm{reco}} - p^{\mathrm{true}})/p^{\mathrm{true}} \rangle$	RMS resolution	Reco fraction
0.080–0.113	+131%	73%	3.4%
0.113–0.150	+66%	56%	3.4%
0.150–0.175	+30%	51%	4.3%
0.175–0.250 (threshold)	**−8%**	**31%**	**10.0%**
0.250–0.400	−30%	21%	10.4%

At the threshold the resolution is 31%; in the $[0.113, 0.175]\,\mathrm{GeV}/c$ region it is 51–56% with a +30–66% bias and only $\sim$ 3–4% reconstructability. Extending the measurement to the ν_e threshold would thus add bins carrying large, strongly model-dependent unfolding systematics. For consistency with the ν_e measurement the threshold could be lowered to $0.113\,\mathrm{GeV}/c$ with the $[0.113, 0.175]\,\mathrm{GeV}/c$ region kept in a single coarse bin to contain those systematics, or both phase spaces reported; this is a systematics-versus-cross-analysis-consistency trade rather than a reconstruction limitation.

The low-momentum bin, added. We have implemented this extension for p_π: a $[0.113, 0.175]\,\mathrm{GeV}/c$ bin is prepended to the pion-momentum binning, lowering the measured phase space to the ν_e threshold. The truth-level signal below $0.175\,\mathrm{GeV}/c$ is recovered from the stored signal-component branches (the reconstruction reproduces `MC_Signal` above $0.175\,\mathrm{GeV}/c$ to 0.4%, the residual being the unstored kaon veto); 3323 true signal events populate the new bin. Its selection efficiency is 2.2% — an order of magnitude below the 15–17% of the standard bins (Table 6), as anticipated from the reconstruction turn-on. Despite this, the Wiener–SVD extraction remains well behaved with the bin included: the M_A^{RES} fake-data closure gives $\chi^2/\mathrm{ndf} = 1.41/6$ ($p = 0.97$) and the flux term is unchanged (17% reco, 18.4% unfolded, amplification 1.08×). The bin therefore adds to the measurement, at the price of a large per-bin uncertainty driven by the low efficiency.

The generator predictions have been regenerated on the six-bin scheme (the ppi histogram uses $p_\pi > 0.113$ while the other observables retain $p_\pi > 0.175$, mirroring the analysis). GENIE, NuWro and NEUT all provide the low bin; GiBUU could not be re-binned as its generator-level event record was not retained, so it is omitted from the extended p_π comparison. Against the (fake) data the generators give $\chi^2/\mathrm{ndf} = 11.1/6$ (GENIE, $p = 0.09$), $7.0/6$ (NuWro, $p = 0.33$) and $6.3/6$ (NEUT, $p = 0.39$): NEUT and NuWro describe the extended spectrum well, while GENIE is mildly disfavoured, consistent with the known GENIE over-prediction of low-momentum $\mathrm{CC}\pi^+$ production.

Table 6: Selection efficiency per true p_π bin with the low-momentum bin added. The new $[0.113, 0.175]\,\mathrm{GeV}/c$ bin sits at 2.2%, an order of magnitude below the standard bins.

True p_π [GeV/c]	N_{true}	N_{reco}	Efficiency
0.113–0.175 (new)	3323	74	**2.2**%
0.175–0.250	3368	526	15.6%
0.250–0.320	2190	367	16.8%
0.320–0.420	2831	479	16.9%
0.420–0.550	2499	410	16.4%
0.550–1.000	2933	429	14.6%

5 Fiducial Volume

The fiducial volume (FV) is defined in `selection/FV_new.h`:

$$x \in [10.0,\ 246.0]\,\text{cm}, \quad y \in [-101.0,\ +101.0]\,\text{cm}, \quad z \in [10.0,\ 986.0]\,\text{cm}, \tag{5}$$

with the dead-wire region $z \in [675.1,\ 775.1]$ cm excluded. The total FV volume is

$$\begin{aligned} V_{\text{FV}} &= (246 - 10) \times 202 \times (976 - 100)\,\text{cm}^3 \\ &= 236 \times 202 \times 876\,\text{cm}^3 \approx 4.17 \times 10^7\,\text{cm}^3. \end{aligned} \tag{6}$$

A wider *containment* region (2 cm inset from the physical TPC walls) is used to require that the pion candidate track endpoint is contained:

$$x \in [2.0,\ 254.35]\,\text{cm}, \quad y \in [-113.53,\ +107.47]\,\text{cm}, \quad z \in [2.1,\ 1034.9]\,\text{cm}. \tag{7}$$

6 Event Selection

The event selection is implemented in the framework selection class `src/selections/CC1mu1piXp.cxx` (which produces the response matrices and spectra for all results in this note); the standalone `selection/ccpi_selection.C` pipeline used for the cross-pipeline validation of Appendix B implements the byte-identical cut sequence. The selection comprises 11 sequential cuts.

6.1 BDT-based particle identification

Three gradient-boosted decision trees (TMVA), trained on the MicroBooNE overlay MC, supply the calorimetric particle identification used in the muon- and pion-candidate cuts. All three share a common set of per-track inputs: the three Bragg-peak dE/dx likelihoods under the proton, muon and MIP hypotheses (`trk_bragg_{p,mu,mip}_v`), the log-likelihood-ratio PID score (`trk_llr_pid_score_v`), the Pandora track-vs-shower score (`trk_score_v`), and the space-charge-corrected track end-point coordinates (`trk_sce_end_{x,y,z}_v`):

- **MIP BDT** (`bdt_mip`; weights `dataset_MIP_BDT_no_len`): separates minimum-ionising tracks (muons, pions) from stopping protons. Both the muon and the pion candidate are required to have `bdt_mip` > -0.1 (Cuts 4–5).
- **Pion BDT** (`bdt_pi`; weights `dataset_pion_BDT_no_len`): a dedicated pion-versus-proton discriminant applied to the pion candidate, `bdt_pi` > -0.1 (Cut 5).
- **Muon BDT** (weights `dataset_muon_BDT`): the same inputs plus the track length (`trk_len_v`). Rather than a threshold, it *ranks* the muon-candidate tracks — the highest-scoring track is chosen as the muon.

The two candidate-PID BDTs deliberately omit track length so the score does not bias the momentum spectra. The weight files are resolved from `$CC1MU1PIXP_BDT_WEIGHTS_DIR`, falling back to the in-tree `booster_decision_tree/` copy. A fourth score, `bdt_cosmic`, is available for cosmic rejection but is disabled by default (no benefit in an A/B test; the Pandora topological score already provides the cosmic rejection at Cut 3).

6.2 Cut definitions

Cut 0. EventsInTrueFV. All events. Reference denominator for truth-level studies.

Cut 1. NeutrinoSlice. ≥ 1 reconstructed track (`ntracks > 0`), `bdt_cosmic > 0.0` (currently disabled; no benefit found in A/B test), and `swtrig == 1`.

Cut 2. InFiducialVol. The SCE-corrected reconstructed neutrino vertex lies inside the FV.

Cut 3. Topological. `topological_score > 0.67`. This Pandora-produced multivariate score distinguishes cosmic-ray topologies from neutrino-like topologies and provides the primary cosmic rejection: EXT events are reduced by a factor of $\sim$33 between Cut 2 and Cut 3.

Cut 4. MuonCandidate. At least one primary PFP track satisfies the muon PID criteria of Table 7. The longest qualifying track is the muon candidate.

Cut 5. ContainedPion. Exactly one primary track (other than the muon candidate) satisfies the pion PID criteria of Table 8 and has its endpoint inside the containment volume. The track with the highest LLR PID score is the pion candidate.

Cut 6. MuonIn3Planes. The muon candidate has ≥ 1 hit on each of the U, V, Y planes.

Cut 7. PionIn3Planes. The pion candidate has ≥ 1 hit on each of the U, V, Y planes.

Cut 8. ShowerCut. Zero primary-level EM showers, unless exactly one shower is present with fewer than 50 Y-plane hits and a vertex distance > 10 cm (consistent with a delta-ray or low-energy nuclear photon).

Cut 9. OpeningAngle. The reconstructed μ–π opening angle satisfies $\theta_{\mu\pi} < 2.6$ rad ($\approx 149°$), rejecting the large-angle cosmic/dirt background. No lower bound is applied (the small-angle region is high-purity). The same cut is imposed on the signal definition (Section 4).

Cut 10.

Nonprotons (final cut). Fewer than 4 primary tracks have LLR PID > 0.1 (non-proton-like), and the total number of primary reconstructed tracks is less than 5.

Table 7: Muon candidate PID thresholds (applied at Cut 4).

Variable	Threshold	Description
`trk_score_v`	> 0.80	Track-like PFP
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 10.0 cm	Minimum track length
LLR PID (`trk_llr_pid_score_v`)	> 0.20	Muon-like
MIP BDT (`bdt_mip`)	≥ -0.10	Minimum-ionising-particle

Table 8: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. `trk_bragg_pion_v`: pion-hypothesis Bragg-peak likelihood score (rejects stopping protons).

Variable	Threshold	Description
LLR PID	> 0.10	Pion / non-proton-like
Vertex distance	< 4.0 cm	Attached to ν vertex
Track length	> 20.0 cm	Minimum track length
MIP BDT (`bdt_mip`)	> -0.10	MIP-like
Pion BDT (`bdt_pi`)	> -0.10	Pion-like (vs. proton)
Bragg-pion score	≥ 0.08	Proton rejection
Endpoint contained	yes	Inside containment volume

6.3 Kinematic reconstruction

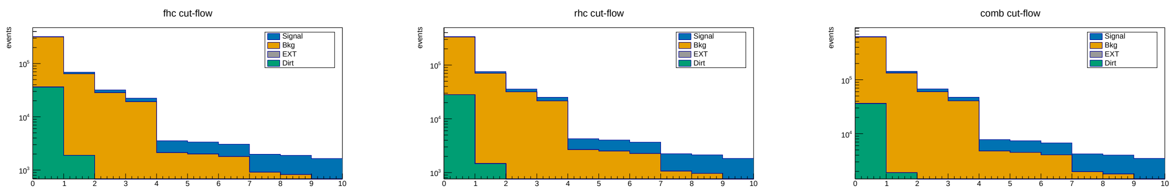
Muon momentum. Range-based momentum from `trk_range_muon_mom_v` (primary estimator); MCS-based momentum from `trk_mcs_muon_mom_v` as cross-check.

Pion momentum. Estimated from the proton-hypothesis range kinetic energy (`trk_energy_proton_v`), which tracks the true pion momentum more accurately than the muon-hypothesis estimator for the short, stopping hadronic tracks typical of low-energy pions.

Angles. Polar angles are defined with respect to the detector z-axis: $\cos\theta_\mu = p_z^\mu/|\mathbf{p}^\mu|$, $\cos\theta_\pi = p_z^\pi/|\mathbf{p}^\pi|$. The opening angle $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$ is computed from reconstructed track directions.

7 Cut-Flow and Selection Performance

Figure 1 shows the full-exposure cut-flow yields for all three configurations as stacked plots, and Table 9 gives the detailed per-category, per-cut breakdown for each.



(a) FHC. (b) RHC. (c) Combined.

Figure 1: Cut-flow yields at *full exposure* for FHC, RHC and combined (log scale; MC-prediction stack: signal blue, background orange, EXT grey, dirt green; colourblind-safe Okabe-Ito), signal and background POT-scaled per mode. Data is blind, so this shows the MC-prediction stack; the detailed per-category breakdown is in Table 9.

Table 9: Cut-flow yields at *full exposure* for FHC, RHC and combined (MC prediction; data is blind, so Pred = Data by construction for the Poisson-thrown fake data). Signal and "Bkg" (all non-signal MC: OOFV, NC, CC$1\pi^0$, CC0π, CC$N\pi$, CCK, CCother) are from the ν_μ overlays, POT-scaled per mode; EXT (beam-off cosmic) is scaled by the data/EXT trigger ratio (FHC 5.51, RHC 6.91, combined 12.41); dirt from the GENIE NuMI dirt overlay. Pred. = Signal + Bkg + EXT + Dirt.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.
FHC (Runs 1,2,4,5; 8.857×10^{20} *POT)*					
None	5 612	279 988	22 690 172	36 142	23 011 914
InFiducialVol	4 666	62 393	968 247	1 890	1 037 197
Topological	3 535	28 198	32 706	204	64 642
MuonCandidate	3 182	19 091	14 423	137	36 833
ContainedPion	1 418	2 101	1 279	25	4 824
MuonIn3Planes	1 358	1 984	1 064	17	4 423
PionIn3Planes	1 264	1 788	810	7.0	3 869
ShowerCut	1 068	907	529	6.2	2 510
OpeningAngle	1 060	823	176	2.0	2 062
Nonprotons	**993**	**653**	**121**	**1.3**	**1 768**
RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} *POT)*					
None	6 139	304 259	28 430 414	28 032	28 768 844
InFiducialVol	5 130	68 998	1 213 197	1 466	1 288 792
Topological	3 969	31 622	40 979	158	76 729
MuonCandidate	3 569	21 446	18 072	106	43 193
ContainedPion	1 580	2 656	1 603	19	5 859
MuonIn3Planes	1 516	2 503	1 333	13	5 365
PionIn3Planes	1 413	2 260	1 016	5.5	4 694
ShowerCut	1 173	1 061	663	4.8	2 902
OpeningAngle	1 165	965	221	1.5	2 353
Nonprotons	**1 084**	**747**	**152**	**1.0**	**1 985**
Combined (1.994×10^{21} *POT)*					
None	11 752	584 247	51 065 438	36 142	51 697 579
InFiducialVol	9 797	131 391	2 179 091	1 890	2 322 169
Topological	7 504	59 820	73 605	204	141 134
MuonCandidate	6 751	40 536	32 460	137	79 884
ContainedPion	2 999	4 757	2 879	25	10 660
MuonIn3Planes	2 874	4 487	2 395	17	9 773
PionIn3Planes	2 677	4 048	1 824	7.0	8 556
ShowerCut	2 241	1 968	1 191	6.2	5 406
OpeningAngle	2 226	1 788	397	2.0	4 412
Nonprotons	**2 077**	**1 400**	**273**	**1.3**	**3 752**

Summary at the final cut (reference phase space, $p_\mu, p_\pi > 0.1\,$GeV/c).

- Signal MC (POT-weighted): 324.7 events
- Total background: $243.4 = 198.5$ (ν-bkg) $+ 40.9$ (EXT) $+ 3.8$ (dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- **Signal purity: 57.2%**
- **Signal efficiency: 12.1–12.6%**
- **Measured phase space (reported result): efficiency** 15.6%**, purity** 54.9% at full Run-1 statistics (selected signal 311.9, generated 1999, background 253.7; Table 10)

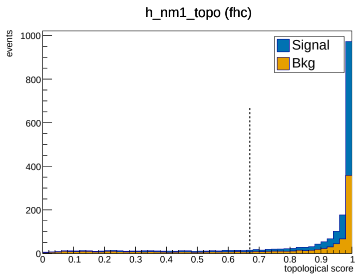
Measured phase space and efficiency turn-on. The reference selection above has near-dead efficiency at low momentum and at the large opening-angle extreme: from the fine truth distributions, the per-bin efficiency is only ~2–5% for $p_\mu, p_\pi < 0.15\,\mathrm{GeV}/c$ (the 10/20 cm track-length cuts) and the region above the reco $\theta_{\mu\pi} = 2.6$ rad cut is dominated by cosmics (~7% purity). The measured phase space $p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad (Section 4) raises and flattens the efficiency to ~13–20% (Fig. 12). Table 10 compares the selection across the iterations. Two lessons: (i) the 2.6 rad *upper* cut is essential — dropping it admits 3.4× more cosmic background and cuts the purity to 45%; (ii) a *lower* opening-angle cut is counterproductive — the small-angle region is high-purity (~58%), so cutting it lowers the selected signal without improving purity, which is why no lower cut is used.

Table 10: Selection performance across the phase-space iterations (NuWro fake data). "no upper cut" drops the $\theta_{\mu\pi} < 2.6$ cut; "$[0.314, 2.6]$" adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

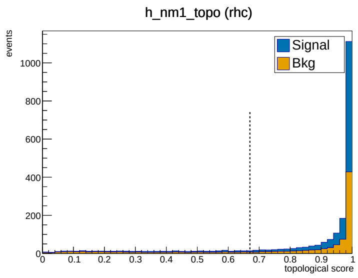
	Reference (p>0.1)	no upper cut (p>.15/.175)	$[0.314, 2.6]$ (lower cut)	**Final** (θ<2.6)
Generated signal	2684	2055	1916	1999
Selected signal	324.7	320.3	301.8	311.9
Efficiency	12.1%	15.6%	15.8%	**15.6%**
EXT (cosmic)	40.9	139.0	38.8	40.9
Total bkg	243.6	385.4	252.3	256.4
Purity	**57.1%**	45.4%	54.5%	**54.9%**
sig × purity	185.6	145.4	164.4	**171.2**

7.1 Key N−1 distributions

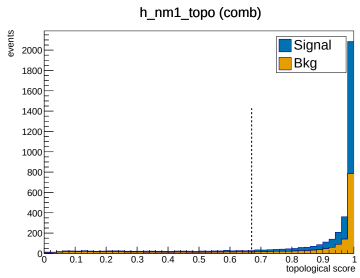
Figure 2 shows the N−1 distributions (all cuts applied except the one under study) for the two cuts instrumented per event — the topological score (Cut 3) and the μ–π opening angle (Cut 9) — for all three configurations. The shapes are consistent across horn modes, as expected since the selection is identical; RHC and combined simply carry the larger exposure. Signal (blue) and background (orange) are stacked and the dashed line marks the applied threshold. The remaining cut variables are shown at the final cut in Sec. 7.3.



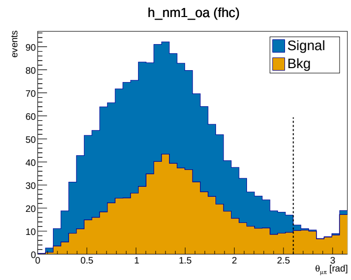
(a) Topo. score, FHC.



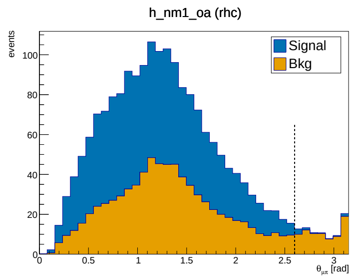
(b) Topo. score, RHC.



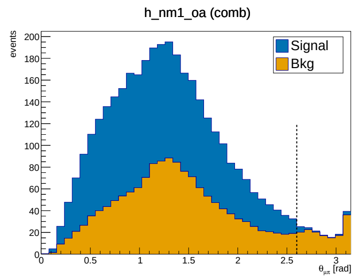
(c) Topo. score, comb.



(d) Opening angle, FHC.



(e) Opening angle, RHC.

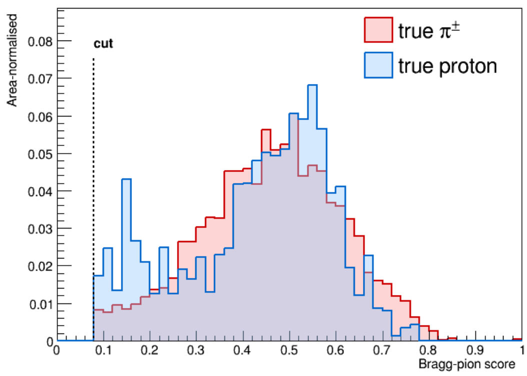


(f) Opening angle, comb.

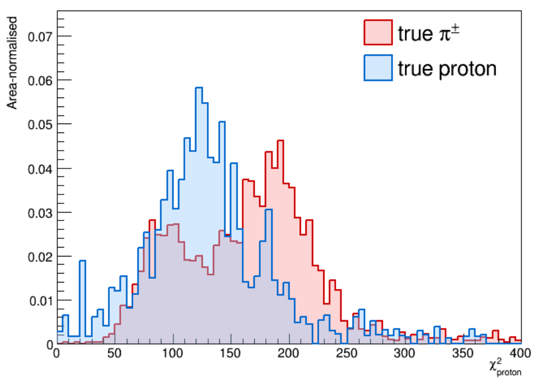
Figure 2: N−1 distributions for the topological score (Cut 3, top) and μ–π opening angle (Cut 9, bottom), for FHC, RHC and combined. Signal (blue) and background (orange) stacked; dashed line = applied threshold. Colourblind-safe Okabe-Ito palette.

7.2 Pion proton-rejection variables

Figure 3 shows the two main proton-rejection variables on the pion candidate, evaluated on MC truth-labelled pions ($\pi^{\pm}$) and protons at the final cut stage, used to tune the working-point thresholds.



(a) Bragg-pion score (`trk_bragg_pion_v`). Working-point threshold: ≥ 0.08.

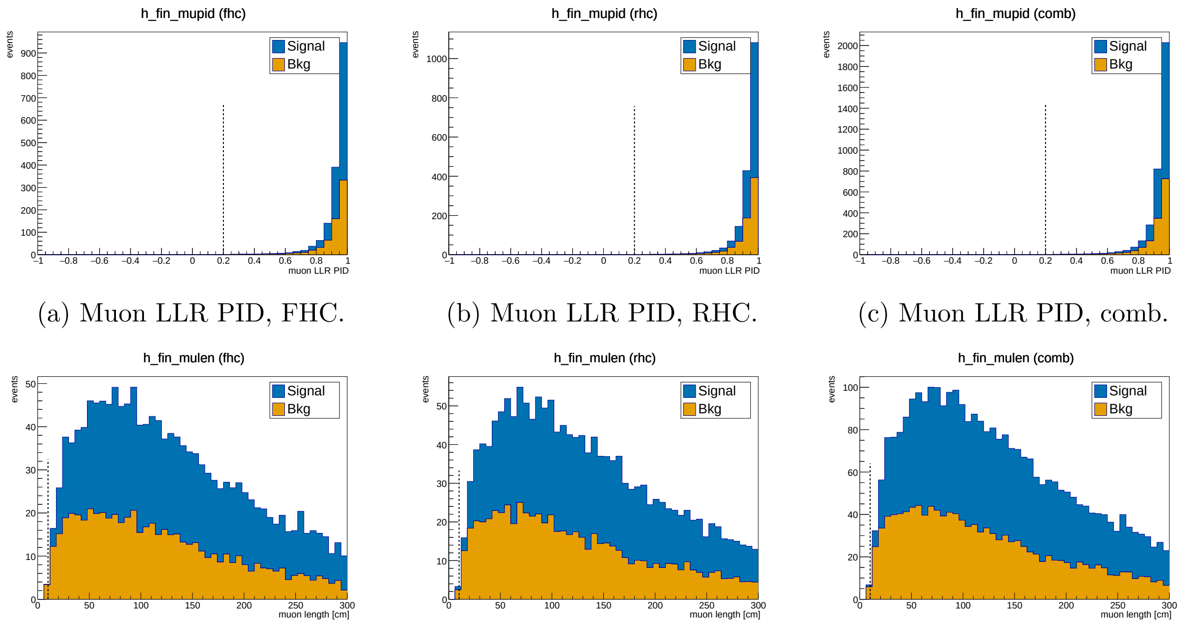


(b) χ^2 proton PID (`trk_pid_chipr_v`).

Figure 3: Pion-candidate PID distributions for true $\pi^{\pm}$ (red) and true protons (blue) at the final cut stage. The Bragg-pion score provides the primary proton rejection; χ^2_p is studied as cross-check.

7.3 Final-cut data/MC distributions

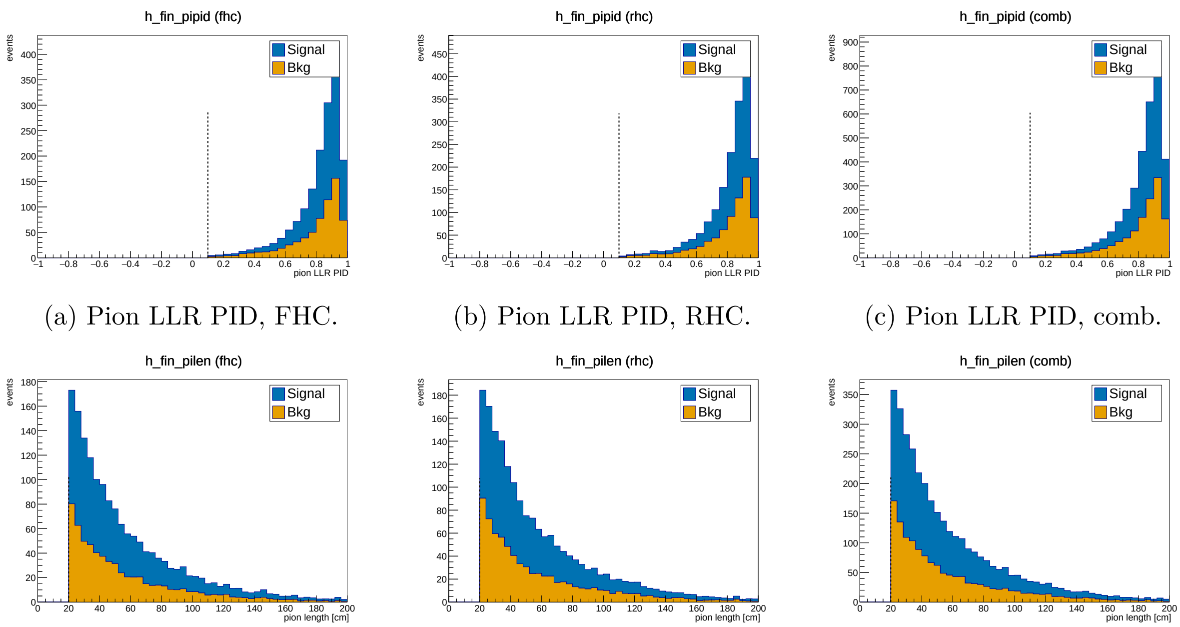
Figures 4 and 5 show the candidate-level distributions at the final selection cut for the four instrumented PID and track-length variables, for all three configurations. Signal (blue) and background (orange) are stacked (colourblind-safe Okabe-Ito palette).



(a) Muon LLR PID, FHC. (b) Muon LLR PID, RHC. (c) Muon LLR PID, comb.

(d) Muon length, FHC. (e) Muon length, RHC. (f) Muon length, comb.

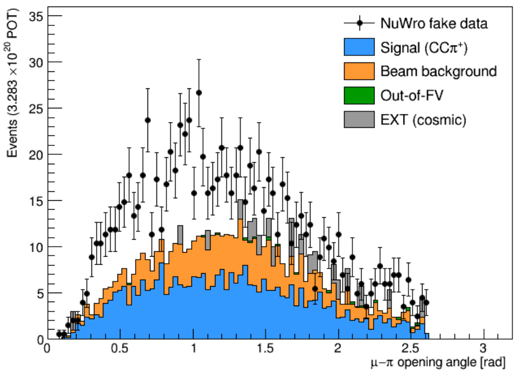
Figure 4: Muon-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC, RHC and combined. Signal (blue) / background (orange) stacked.



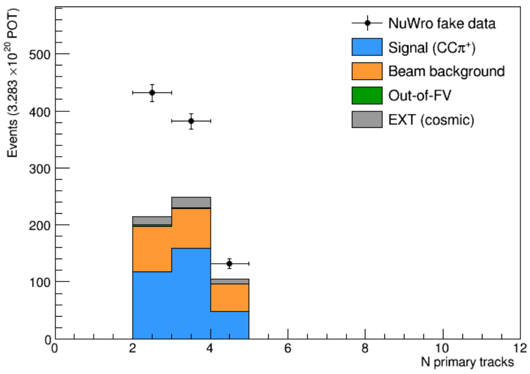
(a) Pion LLR PID, FHC. (b) Pion LLR PID, RHC. (c) Pion LLR PID, comb.

(d) Pion length, FHC. (e) Pion length, RHC. (f) Pion length, comb.

Figure 5: Pion-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC, RHC and combined. Signal (blue) / background (orange) stacked.



(a) μ–π opening angle.



(b) Number of primary tracks.

Figure 6: Data/MC comparison for topological variables at the final cut.

7.4 Background decomposition

Table 11 shows the truth-level background breakdown at the final cut.

Table 11: Background composition at the final cut (truth-level MC categories, POT-weighted).

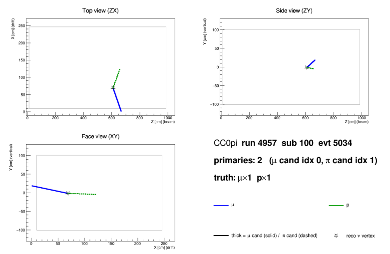
Category	Events	Fraction
CC0π (proton mis-ID as pion)	~ 82	$\sim 41\%$
CC$N\pi$ (≥ 2 pion events)	~ 67	$\sim 34\%$
OOFV (vertex outside FV)	~ 18	$\sim 9\%$
NC (neutral current)	~ 14	$\sim 7\%$
CC1π^0	~ 8	$\sim 4\%$
CC other / CC kaon	~ 6	$\sim 3\%$
ν_e / other flavour	~ 4	$\sim 2\%$
EXT (cosmic)	40.9	—
Dirt	3.8	—

The dominant background ($\sim 41\%$) is CC0π interactions in which a proton is reconstructed as the pion candidate. This mis-identification was verified directly by 2D event displays (truth-labelled by backtracked PDG): the pion candidate track shows a green (proton) truth colour with a stopping Bragg peak, while the genuine muon track is correctly identified.

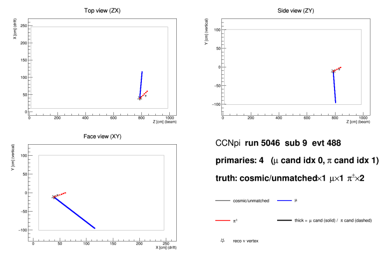
At the current working point (`trk_bragg_pion_v` ≥ 0.08), the pion candidate is truth-matched to a proton in $\approx 18\%$ of cases. Tighter Bragg-score thresholds (0.20–0.30) were studied but shown not to improve the figure of merit: the pion-to-proton ratio in CCπ^+ signal is $\approx 4:1$, so rejecting protons removes genuine pions at nearly the same rate.

7.5 Event displays

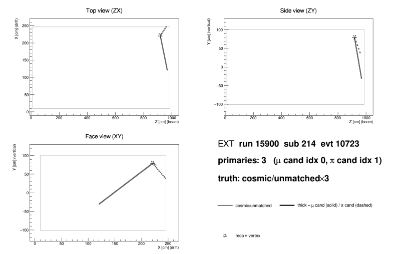
Figure 7 shows representative 2D truth-labelled event displays for the two dominant backgrounds and the EXT cosmic sample. Tracks are coloured by backtracked truth PDG: μ (blue), $\pi^{\pm}$ (red), p (green), cosmic/unmatched (grey). The muon candidate is drawn solid-thick; the pion candidate dashed-thick; the reconstructed ν vertex is shown as a star.



(a) CC0π background: a proton (green, dashed) is reconstructed as the pion candidate.



(b) CC$N\pi$ background: two genuine charged pions present; one is selected as the pion candidate.



(c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 7: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event by `selection/event_display.C`.

8 Unfolding

8.1 Signal extraction

After the event selection, the background-subtracted reco-space signal spectrum for observable x is:

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}, \tag{8}$$

where i runs over reconstructed bins.

8.2 Response matrix

A 2D response matrix A_{ij} (true bin j, reco bin i) is built from selected signal MC using the same binning on both axes. Each column is normalised to unit sum so that $A_{ij} = P(\text{reco in bin } i \,|\, \text{true in bin } j)$. The efficiency vector $\varepsilon_j = \sum_i A_{ij}$ is the ratio of selected (`h_smear_sel`) to generated (`h_true_gen`) signal events per true bin.

8.3 Regularised unfolding: the bias–variance trade-off

Unfolding inverts the folding relation $\vec{m} = A\,\vec{s} + \vec{b}$ to estimate the true signal spectrum $\vec{s}$ from the measured $\vec{m}$. The naive inverse $A^{-1}(\vec{m} - \vec{b})$ is unbiased but useless: bin migration makes A nearly singular, so its inverse amplifies statistical fluctuations into large, anticorrelated bin-to-bin oscillations. Every practical method therefore *regularises* — trading a small, controlled bias for a large reduction in variance. This analysis uses two complementary methods (a fuller pedagogical treatment is given in the companion note `report/unfolding_note.tex`).

Wiener-SVD (primary). Diagonalising the response with an SVD, each mode k is weighted by the Wiener filter $W_k = \sigma_k^2/(\sigma_k^2 + N_k)$ — the minimum-mean-squared-error weighting, $\simeq 1$ for well-measured modes and $\to 0$ for noise-dominated ones — so no regularisation strength is tuned by hand. The price is a *known* linear bias: the unfolded data estimates not $\vec{s}$ but the smeared spectrum $A_C\vec{s}$, where the additional-smearing matrix A_C is published so that every theory prediction is smeared by the same A_C before comparison.

D'Agostini iterative Bayesian (cross-check). Bayes' theorem inverts the migration, $P(C_j|E_i) = R_{ij}\pi_j / \sum_l R_{il}\pi_l$, and the unfolded estimate is $\hat{s}_j = \varepsilon_j^{-1}\sum_i P(C_j|E_i)(m_i - b_i)$. The MC prior π_j is updated with the result and iterated; the *number of iterations* is the regularisation knob (few $\Rightarrow$ prior-biased, many $\Rightarrow$ noisy). It carries no A_C, so its integral is essentially unbiased and theory is compared raw.

Table 12: Trade-offs of the two unfolding methods. Neither is "more correct"; they make complementary choices about how to trade bias for variance and how to expose the residual bias.

	Wiener-SVD	**D'Agostini (iter. Bayes)**
Regularisation	Wiener filter (automatic, from S/N)	Iteration count n (chosen)
Residual bias	Explicit A_C, published	Implicit; set by early stopping
Result integral	Suppressed, observable-dependent	Essentially unbiased (physical)
Theory comparison	Smear theory by A_C	Raw theory
Positivity	Not guaranteed	Guaranteed
Uncertainties	Closed-form, linear	Correlated across iters.; needs care

Wiener-SVD is the primary result — for its objective regularisation and its explicit, reportable bias A_C; D'Agostini serves as the quantitative cross-check (Sec. 8.7), where its flat, iteration-stable integral, a consistent $\sim 20\%$ above Wiener-SVD, confirms that the observable-to-observable spread of the Wiener-SVD integrated cross sections (Table 18) is the A_C smearing rather than a physical or acceptance effect.

8.4 Wiener-SVD unfolding

The primary method is Wiener-SVD [5], implemented in `xsec/WienerSVD.h` (`RunWienerSVD_FW`) as a faithful replica of the official XSecAnalyzer `WienerSVDUnfolder`: Cholesky whitening of the data covariance, regularisation inside the SVD of $R\,C^{-1}$, and the BNL filter (Eq. 3.24 of Ref. [5]). The method applies a Wiener filter in the SVD basis, providing adaptive regularisation whose strength is set by the data covariance. Because the custom pipeline does not throw systematic universes, the covariance is imported from the framework (the prediction covariance scaled to the custom prediction, plus the custom data Poisson; Section A); a stat-only covariance under-regularises and over-corrects the integral by $\sim 26\%$.

The differential cross section is:

$$\left.\frac{d\sigma}{dx}\right|_i = \frac{\hat{N}_i^{\text{unfold}}}{\varepsilon_i \cdot \Delta x_i \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \tag{9}$$

where $\hat{N}_i$ is the Wiener-SVD unfolded signal count in true bin i, $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$ is the total integrated flux, and N_{Ar} is the number of argon target nuclei in the FV. Equivalently, the events-to-cross-section conversion is conv $= N_{\text{Ar}}\,\Phi/10^{-38}$ (so $\sigma_i = \hat{N}_i/(\varepsilon_i\,\text{conv})$), with the authoritative flux constant $\phi = 6.81159 \times 10^{-10}$ cm^{-2}/POT ($E > 60$ MeV; Sec. 2). At the fake-data POT the inputs are:

Quantity	Value
N_{Ar} (FV, dead-region excluded)	8.710×10^{29}
Flux constant ϕ ($E > 60$ MeV)	6.81159×10^{-10} cm^{-2}/POT
POT	7.630913×10^{20}
Integrated flux $\Phi = \phi \times$POT	5.198×10^{11} cm^{-2}
conv $= N_{\text{Ar}}\,\Phi/10^{-38}$	4.527×10^{3}

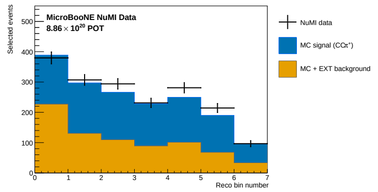
An integrated cross section is computed as the width-weighted sum,

$$\sigma_{\text{int}} = \sum_i \left.\frac{d\sigma}{dx}\right|_i \Delta x_i, \tag{10}$$

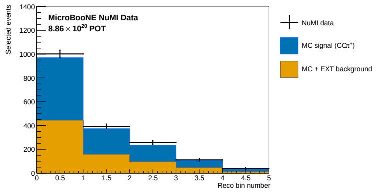
and serves as a closure check: all five observables must agree.

8.5 Reconstructed spectra and response matrices

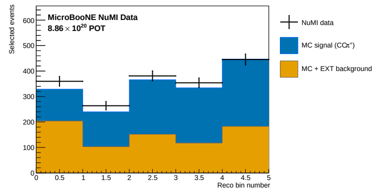
Figure 8 shows the background-subtracted reco-space signal spectra for all five observables, overlaid with the MC signal prediction and the individual background components.



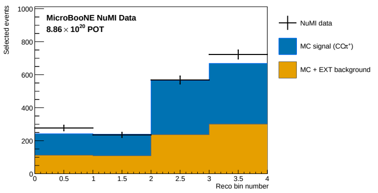
(a) p_μ



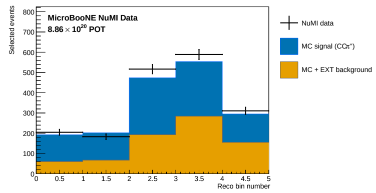
(b) p_π



(c) $\cos\theta_\mu$

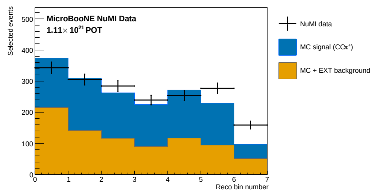


(d) $\cos\theta_\pi$

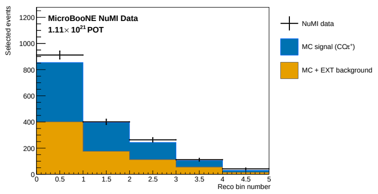


(e) $\theta_{\mu\pi}$

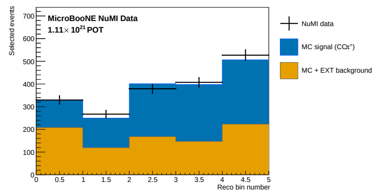
Figure 8: Reconstructed-space spectra at the final cut for all five observables, FHC full exposure (Runs 1+2+4+5): Poisson-thrown central-value fake data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). The fake data is drawn from the central-value Monte Carlo and so agrees with the stacked prediction by construction (a reco-level self-closure); the differential results and their closure against the A_C-smeared truth are shown in Fig. 14.



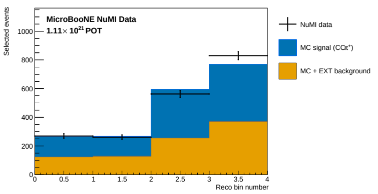
(a) p_μ



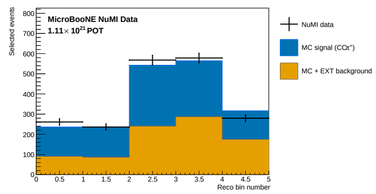
(b) p_π



(c) $\cos\theta_\mu$

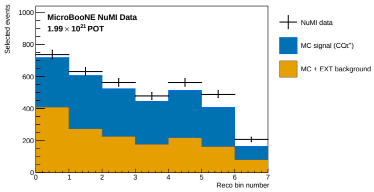


(d) $\cos\theta_\pi$

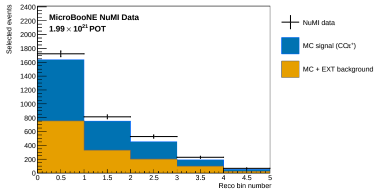


(e) $\theta_{\mu\pi}$

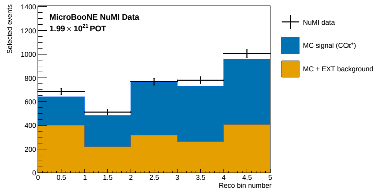
Figure 9: Reconstructed-space spectra, RHC full exposure (Runs 1+2+3+4a+4b+4c); same components as Fig. 8.



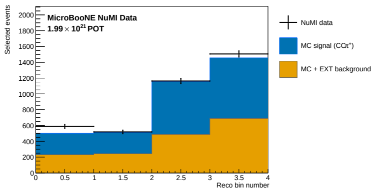
(a) p_μ



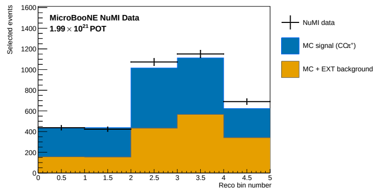
(b) p_π



(c) $\cos\theta_\mu$



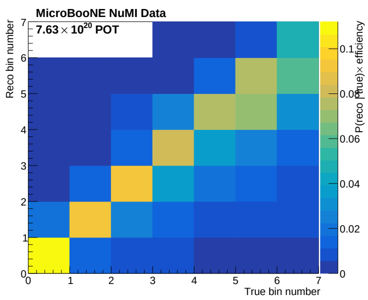
(d) $\cos\theta_\pi$



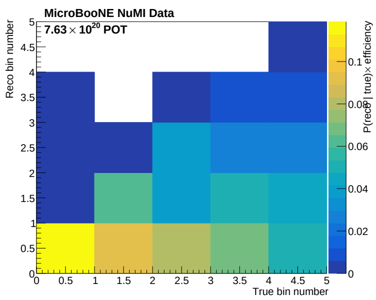
(e) $\theta_{\mu\pi}$

Figure 10: Reconstructed-space spectra, combined FHC+RHC (per-mode normalised); same components as Fig. 8.

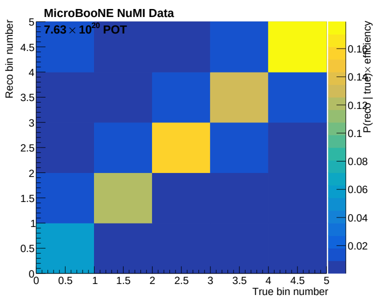
Figure 11 shows the response matrices (normalised per true bin) for the five observables. The diagonal dominance confirms that migration effects are modest for the angular observables; p_μ shows slightly larger migrations in the low-momentum bins.



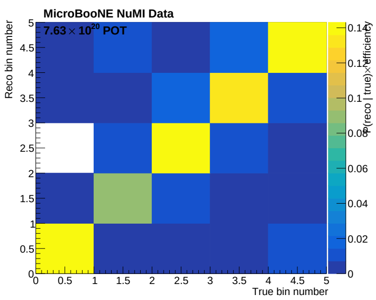
(a) p_μ



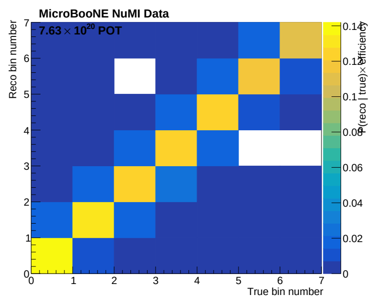
(b) p_π



(c) $\cos\theta_\mu$



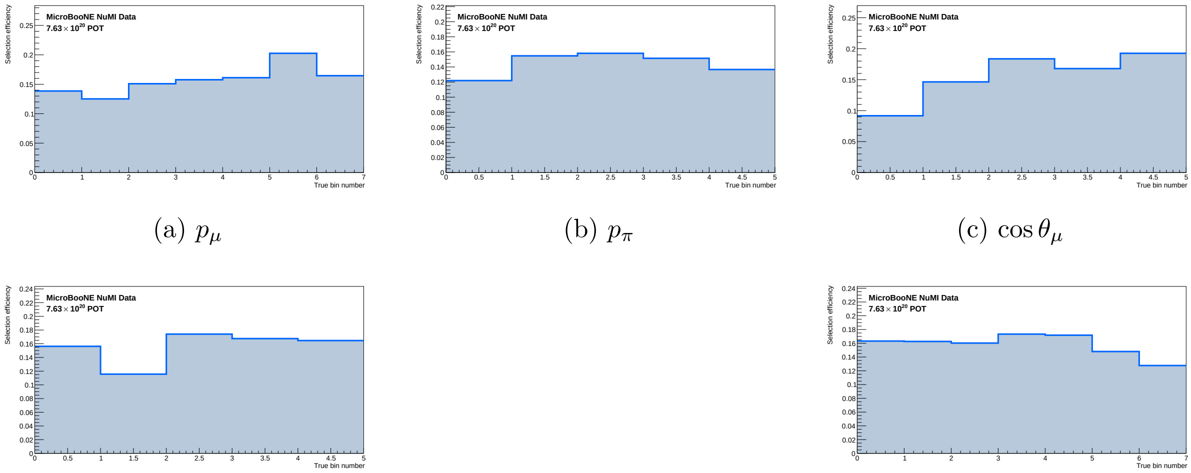
(d) $\cos\theta_\pi$



(e) $\theta_{\mu\pi}$

Figure 11: Response matrices A_{ij} for the five observables. Each column is normalised to unity (migration probability). Colour scale: white = 0, dark blue = 1.

Figure 12 shows the selection efficiency per true bin for all five observables, computed as the ratio of selected to generated signal MC.



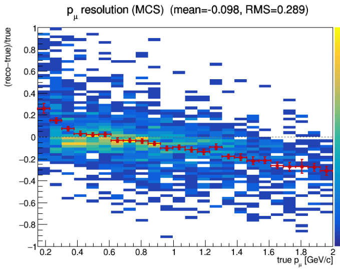
(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

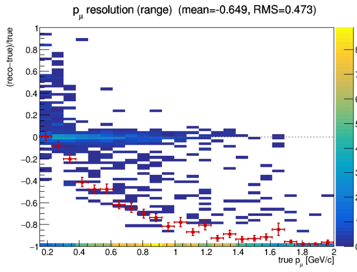
Figure 12: Selection efficiency as a function of true observable value, in the measured phase space ($p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad). Efficiency is the ratio of selected to generated signal MC ($\varepsilon_i = \sum_j A_{ij}$, integrated over reco bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at ∼13–20% across all observables; the former ∼5% low-momentum turn-on bins and the cosmic-dominated large-angle region are removed. The lowest point is the smallest opening-angle bin (∼9% efficiency) — it is retained because it is high-purity (∼58%), not low-purity.

8.6 Reconstruction resolution

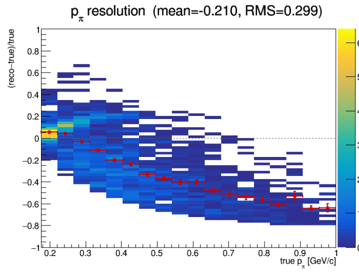
Figure 13 shows the reconstruction resolution of each measured observable, evaluated on selected signal events (GENIE ν_μ overlay, measured phase space). For the momenta the relative resolution (reco − true)/true is shown; for the angular observables the absolute residual (reco − true) is used, since the relative metric diverges as $\cos\theta \to 0$ / $\theta_{\mu\pi} \to 0$. The red points are the mean residual per true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).



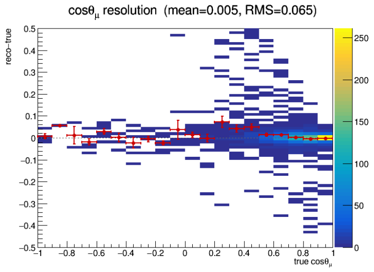
(a) p_μ (MCS): bias -9.8%, RMS 29%.



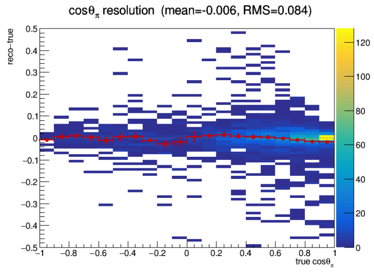
(b) p_μ (range): valid only for contained muons.



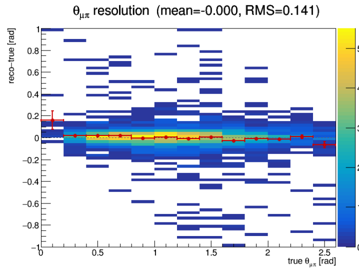
(c) p_π: bias -21%, RMS 30%.



(d) $\cos\theta_\mu$: RMS 0.065.



(e) $\cos\theta_\pi$: RMS 0.084.



(f) $\theta_{\mu\pi}$: RMS 0.14 rad.

Figure 13: Reconstruction resolution vs. true value, selected signal. Momenta: relative (reco − true)/true; angles: absolute (reco − true). Red points: mean residual per true bin. The muon MCS momentum slightly over-estimates at low p_μ and increasingly under-estimates above ∼1 GeV/c; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion momentum (proton-hypothesis range KE) under-estimates true p_π by ∼21%. The angular variables are essentially unbiased with narrow resolution.

8.7 Cross-check methods

Seven additional methods are run in parallel:

Table 13: Unfolding methods.

Label	Description
`wiener_svd`	Wiener-SVD (primary result)
`bin_by_bin`	Efficiency correction only; no migration unfolding
`ibu`	D'Agostini iterative Bayesian, 4 iterations
`tikhonov`	Tikhonov-regularised least-squares, $\tau = 1.0$
`roo_bayes`	RooUnfoldBayes, 4 iterations
`roo_svd`	RooUnfoldSvd
`roo_invert`	RooUnfoldInvert (direct matrix inversion)
`roo_binbybin`	RooUnfoldBinByBin

The D'Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix A_C, its unfolded total is free of the (non-norm-preserving) A_C suppression and is stable against the iteration count. Table 14 compares the two integrated totals for all three configurations: the D'Agostini result sits a consistent ∼24% above the Wiener-SVD total (equivalently, A_C suppresses the quoted Wiener-SVD integral by ∼19%), and it varies by $< 1\%$ across 1, 2 and 4 iterations. For FHC the D'Agostini per-observable truth is flat to 1% (1.111–1.122), directly confirming that the observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 18) is the A_C smearing rather than a physical or acceptance effect.

Table 14: Wiener-SVD vs. D'Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five observables (fake-data truth, 10^{-38} cm^2/Ar). D'Agostini carries no A_C smearing; its ~24% higher total is consistent across configurations and stable to $< 1\%$ over 1/2/4 iterations. The FHC D'Agostini per-observable truth is additionally flat to 1%.

Config	$\langle\sigma\rangle_{\mathrm{WSVD}}$	$\langle\sigma\rangle_{\mathrm{DAg}}$	DAg/WSVD	DAg 1→4 iter
FHC	0.91	1.12	1.23	$< 1\%$
RHC	0.75	0.93	1.24	$< 1\%$
Combined	0.80	1.00	1.25	$< 1\%$

9 Full-exposure results: FHC, RHC, and combined

This is the primary result of the analysis: the CC1μ1$\pi^{\pm}$ cross section extracted over the full available exposure in forward horn current (FHC), reverse horn current (RHC), and their combination, all on Poisson-thrown central-value fake data. The FHC measurement uses Runs 1, 2, 4 and 5 (ν_e-matched data POT 8.857×10^{20}); RHC uses Runs 1, 2, 3, 4a, 4b, 4c (1.108×10^{21}); the combined measurement uses all fourteen overlay files at the joint POT 1.994×10^{21}. All three close cleanly against the A_C-smeared truth for every observable (Tables 15–17); the differential results are shown in Figs. 14–16.

Table 15: Full-exposure FHC ({Run 1,2,4,5}, data POT 8.857×10^{20}, native Run-4 FHC detector variations): reco-level and unfolded flux systematic, unfolding amplification, and fake-data closure χ^2/ndf (with p-value).

Observable	Flux (reco)	Flux (unfolded)	Amplification	Closure χ^2/ndf
p_μ	17.1%	23.9%	1.40×	0.55/7 ($p = 1.00$)
p_π	16.9%	23.9%	1.41×	0.29/5 ($p = 1.00$)
$\cos\theta_\mu$	16.7%	23.9%	1.43×	0.93/5 ($p = 0.97$)
$\cos\theta_\pi$	16.6%	27.7%	1.67×	0.10/4 ($p = 1.00$)
$\theta_{\mu\pi}$	16.8%	22.3%	1.33×	1.84/5 ($p = 0.87$)

We extend the measurement to the reverse-horn-current beam as both a standalone measurement and a combined $\nu_\mu + \bar{\nu}_\mu$ result. The RHC *flux* is $\bar{\nu}_\mu$-dominant (54.6% $\bar{\nu}_\mu$, versus FHC's 35%), with authoritative >60 MeV integrals $\Phi^{\mathrm{RHC}}_{\nu_\mu} = 2.923\times10^{-10}$, $\Phi^{\mathrm{RHC}}_{\bar{\nu}_\mu} = 3.523\times10^{-10}$, total 6.446×10^{-10} cm^{-2}POT^{-1} (the FHC-convention computation reproduces the authoritative FHC values exactly). The *measured event rate*, however, remains ν_μ-dominant (76% ν_μ, 20% $\bar{\nu}_\mu$) because the ν_μ cross section is ~2–3× larger than the $\bar{\nu}_\mu$ one. The RHC overlays (Runs 1, 2, 3, 4a, 4b, 4c) are all verified as genuine ν_μ overlays and processed — 4.05×10^6 events, combined MC POT 1.546×10^{22}, mutually consistent (2.62×10^{-16} events/POT) with the PPFX weights — covering the full RHC exposure (Run 3, the largest period at 5.52×10^{21} MC POT / 1.44×10^6 events, was added last). Native Run-4 RHC detector-variation samples (the standard eight-knob set: light-yield down/Rayleigh, recombination, space charge, and the four wire-modification variations) are now used for the RHC detector systematic, verified as genuine ν_μ overlays with a uniform 3.80×10^{-16} events/POT across all knobs; they replace the Run-1 FHC stand-in used previously. Native RHC dirt is still taken from FHC. The standalone result below uses Poisson-thrown central-value fake data at the ν_e-matched RHC data POT 1.108×10^{21}. The generator predictions are re-averaged to the RHC flux composition; the combined FHC+RHC predictions are the fluence-weighted combination of the two horn modes (0.458 FHC / 0.542 RHC, each mode weighted by $\Phi_{\rm tot}\times$POT).

The combined FHC+RHC measurement uses the summed 14-file response over the full exposure (5.6×10^6 events, combined data POT $D_{\rm comb} = 1.994 \times 10^{21} = D_{\rm FHC} + D_{\rm RHC}$), with each horn mode normalised to its *own* data exposure. This is a genuine subtlety: the detector-variation covariance is absolute (POT-scaled reco counts), so a single POT scale applied to all files would weight the modes by their Monte-Carlo POT (0.62:1 FHC:RHC) rather than by data exposure (0.80:1). Each mode's numuMC and detVar samples therefore carry a per-mode `summed_pot` (FHC 2.158×10^{22}, RHC 2.782×10^{22}, $= S_{\rm mode} \times D_{\rm comb}/D_{\rm mode}$) so the framework's $D_{\rm comb}$/`summed_pot` scaling reduces to $D_{\rm FHC}/S_{\rm FHC}$ and $D_{\rm RHC}/S_{\rm RHC}$; the fake data is Poisson-thrown with the matching per-mode rate.

Contrary to an earlier concern, the joint *flux systematic* is correct as built. All fourteen numuMC files carry the *same* 600 PPFX multisim universes, which the universe maker accumulates index-matched across both horn modes in a single chain, so universe u is the same hadron-production draw in FHC and RHC. Per-universe summing therefore propagates one parameter draw through both beam configurations and lets each mode's flux response set the result, capturing the physical (not unit) cross-horn correlation automatically — this is the rigorously correct multisim treatment, superior to any hand-built block covariance. The joint result was in fact blocked by a different, purely technical issue: a framework guard that forbade a second detector-variation file of the same type (needed to carry Run-4 FHC *and* Run-4 RHC as `detVarCV`). That guard now accumulates the per-mode detVar samples exactly as it already did for the alternate-CV samples, matching its own in-code TODO. With both in place the combined closure passes for every observable (Table 17), with an unfolded flux systematic of 22–27% that sits between the standalone FHC and RHC values, as expected for the fluence-weighted mix.

The standalone RHC extraction (native Run-4 RHC detector variations) closes cleanly for every observable (Table 16): the A_C-smeared truth is recovered with $\chi^2/\text{ndf} < 0.3$ and $p > 0.94$ throughout. Two features stand out. First, the reco-level flux systematic is $\sim 15\%$, genuinely below the FHC $\sim 17\%$ floor — the $\bar{\nu}_\mu$-dominant flux carries a smaller PPFX hadron-production uncertainty. Second, the unfolding amplification (1.29–1.99$\times$) mirrors FHC, so the unfolded flux term (19.5–28%) is comparable, again data-statistics limited.

Table 16: Standalone RHC ($\bar{\nu}_\mu$-dominant) extraction, full exposure 1.108×10^{21} with native Run-4 RHC detector variations: reco-level and unfolded flux systematic, unfolding amplification, and fake-data closure χ^2/ndf (with p-value).

Observable	Flux (reco)	Flux (unfolded)	Amplification	Closure χ^2/ndf
p_μ	15.4%	27.5%	1.78$\times$	1.37/7 ($p = 0.99$)
p_π	15.8%	23.6%	1.49$\times$	0.21/5 ($p = 1.00$)
$\cos\theta_\mu$	15.0%	26.5%	1.76$\times$	0.43/5 ($p = 0.99$)
$\cos\theta_\pi$	14.9%	28.5%	1.91$\times$	1.03/4 ($p = 0.91$)
$\theta_{\mu\pi}$	15.1%	20.4%	1.35$\times$	0.34/5 ($p = 1.00$)

Table 17: Combined FHC+RHC (ν_μ+$\bar{\nu}_\mu$) extraction over the full joint exposure $D_{\rm comb} = 1.994\times 10^{21}$, per-mode normalised, with native Run-4 FHC+RHC detector variations (accumulated per mode): reco-level and unfolded flux systematic, unfolding amplification, and fake-data closure χ^2/ndf (with p-value). The unfolded flux term (22–27%) lies between the standalone FHC and RHC values.

Observable	Flux (reco)	Flux (unfolded)	Amplification	Closure χ^2/ndf
p_μ	16.2%	22.5%	1.39×	0.58/7 ($p = 1.00$)
p_π	16.3%	23.9%	1.47×	1.27/5 ($p = 0.94$)
$\cos\theta_\mu$	15.8%	25.0%	1.58×	0.28/5 ($p = 1.00$)
$\cos\theta_\pi$	15.7%	27.0%	1.72×	1.19/4 ($p = 0.88$)
$\theta_{\mu\pi}$	15.9%	26.0%	1.64×	0.43/5 ($p = 0.99$)

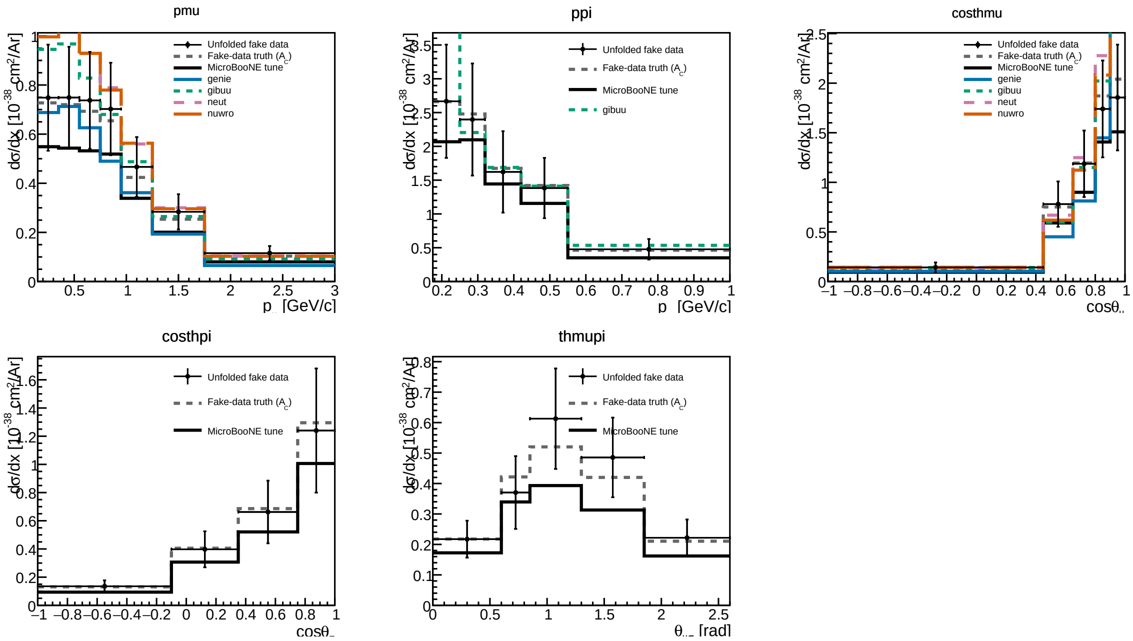


Figure 14: Differential cross sections $d\sigma/dx$ for all five observables on the current Poisson-thrown central-value fake data, FHC full exposure: unfolded fake data (black points), A_C-smeared fake-data truth (grey dashed), the MicroBooNE tune (black solid), and the four generator predictions (GENIE, GiBUU, NEUT, NuWro). The unfolded points recover the injected truth for every observable, the same closure summarised in Tables 16–17.

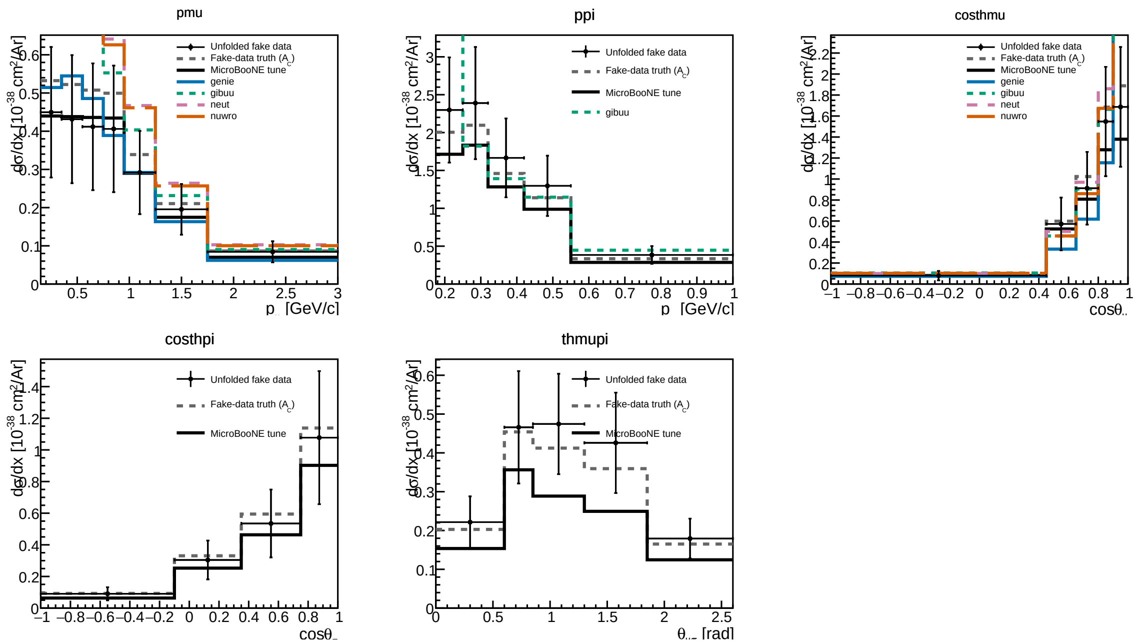


Figure 15: Standalone RHC ($\bar{\nu}_\mu$-dominant) differential cross sections on the current Poisson-thrown fake data, with the RHC-flux-averaged generator predictions. Same layout as Fig. 14.

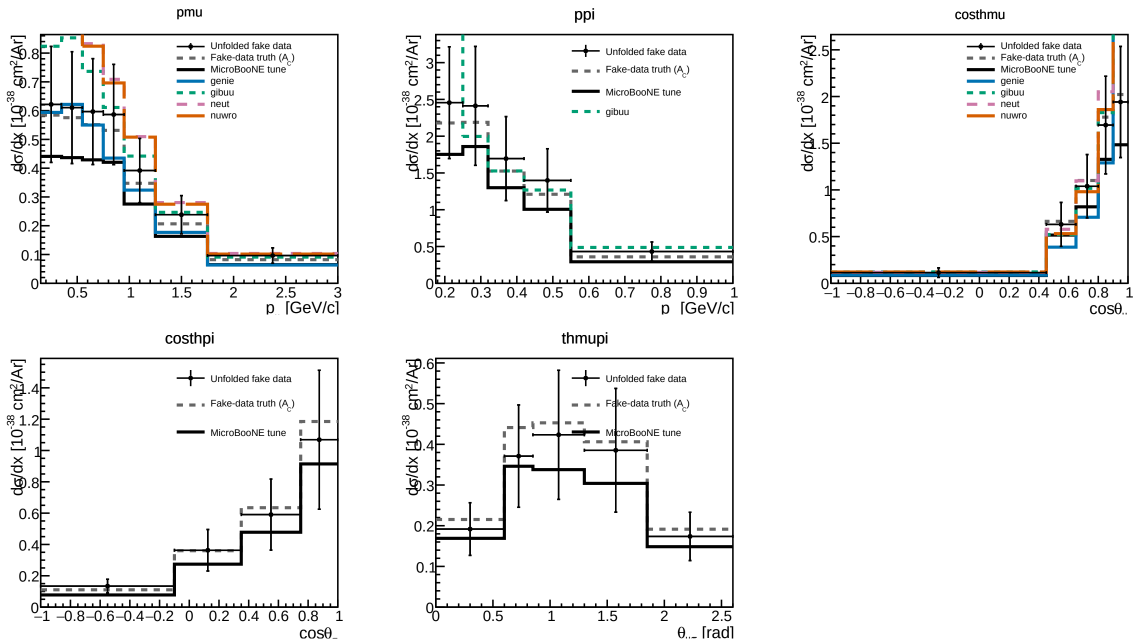


Figure 16: Combined FHC+RHC ($\nu_\mu + \bar{\nu}_\mu$) differential cross sections on the current per-mode-normalised fake data, with the fluence-weighted combined generator predictions.

Table 18: Integrated (unfolded) cross section $\sigma_{\rm int}$ over the measured phase space, for each observable and configuration (current fake data, coarsened $\cos\theta_\pi/\theta_{\mu\pi}$ binning, per-configuration flux; 10^{-38} cm^2/Ar). The value differs between observables because the Wiener-SVD additional-smearing A_C is not norm-preserving; the D'Agostini cross-check (Sec. 8.7) recovers a flatter, $\sim$20% higher total, confirming the spread is a smearing effect. RHC sits below FHC because the RHC beam is $\bar{\nu}_\mu$-dominant (lower flux-averaged cross section).

Observable	FHC	RHC	Combined
p_μ	1.014	0.630	0.840
p_π	0.925	0.848	0.899
$\cos\theta_\mu$	0.900	0.691	0.811
$\cos\theta_\pi$	0.876	0.702	0.788
$\theta_{\mu\pi}$	0.932	0.832	0.741

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 19–21. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the fake-data POT of each configuration. The RHC detector term is the largest single-configuration entry, driven by the $\cos\theta$ observables at the current statistics.

Table 19: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	26.6	33.4	29.9	33.1	28.2
Cross section (GENIE)	7.3	8.9	7.3	9.1	8.8
Flux (PPFX)	22.6	27.0	23.5	27.7	22.3
Detector	6.7	10.0	10.3	10.0	7.5
Reinteraction	3.3	4.1	3.0	4.2	3.4
MC stat	2.3	3.2	3.1	2.4	3.2
EXT stat	4.3	5.7	6.0	5.2	4.1
Data stat	7.4	10.8	10.4	8.0	10.7
POT + targets	2.7	3.5	3.0	3.5	2.8

Table 20: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	37.4	30.7	41.1	41.3	29.5
Cross section (GENIE)	10.1	8.4	9.4	10.2	7.7
Flux (PPFX)	27.5	23.6	26.5	28.5	20.4
Detector	18.2	14.2	25.0	24.7	15.4
Reinteraction	3.9	2.8	3.5	4.4	2.7
MC stat	3.0	2.1	3.3	2.2	2.5
EXT stat	6.8	5.2	8.9	6.2	5.9
Data stat	11.0	7.9	12.3	8.1	9.6
POT + targets	3.8	3.3	3.6	3.9	2.8

Table 21: Combined FHC+RHC systematic breakdown by source, bin-averaged fractional uncertainty (%).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	29.8	31.8	35.2	37.3	35.7
Cross section (GENIE)	8.2	8.5	8.1	9.1	9.0
Flux (PPFX)	22.5	23.9	25.0	27.0	26.0
Detector	14.5	16.0	19.4	21.7	18.4
Reinteraction	3.2	4.6	3.2	4.1	3.6
MC stat	1.9	2.0	2.6	1.6	2.5
EXT stat	5.7	4.3	7.5	5.4	7.7
Data stat	6.7	6.7	8.9	5.5	8.9
POT + targets	2.8	3.2	3.3	3.6	3.4

10 Systematic Uncertainties

Full systematic uncertainty propagation has not yet been performed. Table 22 lists the sources and their estimated impacts.

Table 22: Systematic uncertainty sources. “Not quantified” means the branch is readable but propagation code has not been written.

Source	Branch / method	Impact
Flux (PPFX multisims)	`weightsFlux`	$\sim$ 10–15% norm.
GENIE cross sections	`weightsGenie`	$\sim$ 20–30% (model)
Hadron re-interaction	`weightsReint`	$\sim$ 5–10%
Detector response	Dedicated samples	$\sim$ 5–15%
EXT normalisation	Trigger-count ratio	$\sim$ 3–5%
POT counting	Beam toroids	$\sim$ 2%
Flux integral (absolute)	PPFX integral	Blocks absolute σ
Statistical (427 data events)	—	$\sim$ 5% on σ_{int}

11 Known Issues and Required Follow-up

Table 23: Outstanding issues and their impact on the final result.

Issue	Status	Impact
`FLUX_PER_POT` confirmed	**Resolved**	$\Phi = 6.81159 \times 10^{-10}$ cm^{-2} POT^{-1} from NuMI flux histograms
Unfolding covariance (stat-only → full)	**Resolved**	Framework `PredTotal`+data Poisson imported; removed $\sim 26\%$ bias
EXT subtraction on NuWro fake data	**Resolved**	Cancelled via `fake_data_ext_cancel`; re-enable for real data
$\theta_{\mu\pi}$ binning (10 vs 9, C4)	**Resolved**	Custom aligned to framework 9-bin scheme
Multisim systematics not propagated (custom)	Not started	Imported from framework for the covariance; full custom propagation pending
Negative tail bins	**Resolved**	All bins now positive after EXT cancellation; no longer present
Runs 2, 4, 5 beam-on files not available	Data files needed	$\times 5.4$ exposure increase for the real-data measurement
Generator comparisons (GENIE, NuWro)	Blocked by missing AVX on compute nodes	NEUT comparison runnable locally
CRT veto branches (`crthitpe`, `crtveto`)	Dead (all-zero in NuMI ntuples)	Replaced by `bdt_cosmic` (disabled at default 0.0)
Official XSecAnalyzer NuWro closure	**Done** (all 5 obs)	Cross-validates the unfolding; NuWro $\approx 2\times$ GENIE (Sec. B)
Native RHC detector variations	**Resolved**	Run-4 RHC detVar (8 knobs) now used; replaces the Run-1 FHC stand-in (Sec. 9)
Run-5 FHC detVar Recomb2/SCE	**Resolved**	Were intrinsic-ν_e overlays (0% ν_μ); Run-4 FHC used as the stand-in for those two knobs
Combined FHC+RHC joint flux systematic	**Resolved**	Per-universe summing over the shared 600 PPFX universes is correct; the block was a detVar duplicate-file guard, now fixed to accumulate (Sec. 9)
Combined per-mode normalisation	**Resolved**	Per-mode `summed_pot` so each horn scales to its own data exposure, not MC POT (Sec. 9)

12 Analysis Pipeline

The primary measurement is produced by a single driver, `run_analysis.sh`, which chains the selection, the per-observable cross-section extraction, and the comparison/diagnostic stages:

```
selection/ccpi_selection.C   ->  histograms.root + unfolding_inputs.root
                                  | (5 observables x 8 methods)
xsec/ccpi_xsec.C(obs,method) ->  xsec_{obs}_{method}.root  (40 files)
xsec/compare_unfolding.C     ->  compare_plots/
```

```
xsec/compare_generators.C     ->  gen_compare/
selection/event_display.C     ->  event_displays/
```

Each output file is validated for freshness and non-zero content after the step that produces it. The official-framework cross-check of Section B is driven separately (`xsec_analyzer/run_nuwro_observables.`

13 Binning Optimisation, Systematic Breakdown, and Threshold Validation

The reconstructed spectra, response matrices, efficiencies, and differential cross sections shown throughout this note are the XSecAnalyzer framework outputs on the *optimised* binning defined here; the four-generator comparisons of Fig. 18 onward likewise use it.

13.1 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC ν_μ $1\mu1\pi^\pm$ topology with the momentum/angle cuts removed):

Threshold	below	above	turn-on
$p_\mu > 0.15$ GeV/c	$\sim$0–1.5%	$\sim$9–12%	0.145→0.155
$p_\pi > 0.175$ GeV/c	$\sim$0–3%	$\sim$12–16%	0.165→0.185

Both are correctly placed — lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco−true ≈ 0 at threshold, -0.15 GeV/c at 0.40) from the μ-vs-π mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the p_π reco bins.

13.2 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold ≥ 45 selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

Observable	bins	Total before	Total after
p_μ	22→7	33.2%	**25.0%**
p_π	5→5	27.7%	27.0%
$\cos\theta_\mu$	12→5	31.3%	**26.7%**
$\cos\theta_\pi$	12→5	48.5%	**33.9%**
$\theta_{\mu\pi}$	9→7	28.4%	**23.5%**

13.3 Systematic breakdown by source

The framework builds a covariance matrix for every systematic; the bin-averaged fractional uncertainty from each source group (on the optimised binning) is:

Table 24: Bin-averaged fractional systematic uncertainty (%) by source group on the optimised binning.

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	27.2	28.3	28.8	33.8	25.1
Cross section (GENIE)	8.6	8.5	9.1	9.5	8.1
Flux — PPFX	21.8	23.6	20.7	25.4	18.7
Flux — beamline geom.	2.3	2.2	2.3	3.3	1.6
Detector	9.6	10.2	14.0	14.6	10.5
Reinteraction	3.2	2.6	3.4	3.4	2.5
MC + data stats	8.6	6.5	9.4	11.9	9.3
POT + targets	2.7	2.9	2.7	3.0	2.4

The NuMI flux uncertainty has two parts, listed separately above: the PPFX hadron-production term (the dominant systematic, 19–26%) and the beamline-geometry term. The latter is built from the 12 geometry-variation weights (`Horn_2kA`, `Horn1/2_x/y_3mm`, `Beam_spot_*`, `Beam_shift_*`, `Horns_*_water`, `Target_z_7mm`) injected into the numuMC overlay via `AddBeamlineGeometryWeights`, summed in quadrature as two-sided unisim variations. It is sub-dominant, 1.6–3.3% across the observables. (Building it required a framework fix: the covariance code dereferenced a null histogram when a systematic — here the beamline weights, present in the numuMC but not the dirt overlay — was absent from some MC sample; it now falls back to that sample's central value, i.e. unit weight.) The per-bin breakdowns are shown in Fig. 17.

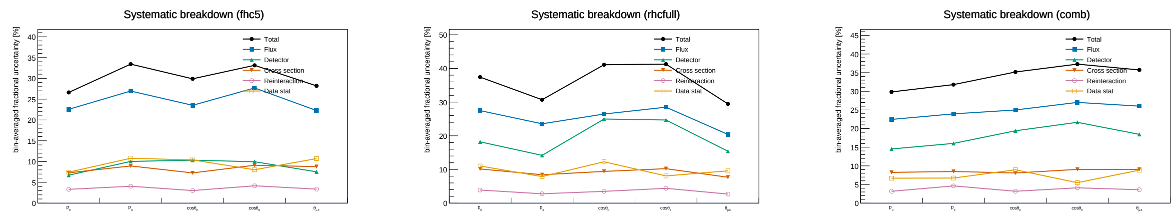


Figure 17: Bin-averaged fractional systematic uncertainty by source vs. observable, for FHC (left), RHC (centre) and combined (right) — the figure form of Tables 19–21. Colours use the Okabe-Ito colourblind-safe palette with distinct markers per source: total (black), flux (blue), detector (green), cross section (vermillion), reinteraction (purple), data stat (orange).

13.4 Anatomy of the flux term: genuine flux vs. unfolding amplification

The flux (PPFX) term dominates the budget, and at 17–26% it is larger than the $\sim 17\%$ normalisation uncertainty usually quoted for the NuMI flux. It is therefore worth asking how much of it is the flux model itself and how much is statistical amplification introduced by the unfolding. To separate the two we evaluate the flux covariance at two stages — on the reconstructed spectrum (before unfolding) and on the unfolded result — and, at each stage, both integrated (the fully correlated, normalisation-like number, from the sum of all covariance elements) and bin-averaged (the mean of the diagonal fractional errors, i.e. the number quoted in Table 24 and Fig. 17).

Table 25: Flux (PPFX) fractional uncertainty before and after unfolding. The reconstructed-level value is $\sim 17\%$ and flat across observables — the genuine flux-normalisation uncertainty. The unfolded value is that same $\sim 17\%$ amplified by the Wiener–SVD propagation through the finite-statistics response; the amplification tracks the data+MC statistical quality of each observable.

	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Flux, reco (integrated / norm)	16.9	17.0	16.7	16.7	16.7
Flux, reco (bin-averaged)	17.7	16.4	16.6	17.0	17.0
Flux, unfolded (bin-averaged)	19.8	22.4	19.8	25.5	17.5
Amplification factor	1.12	1.37	1.20	1.49	1.03
Excess from unfolding (abs.)	+2.1	+6.0	+3.2	+8.5	+0.5
Data+MC stats (unfolded)	7.9	6.3	8.4	12.2	8.8

Three points follow. First, the *genuine* flux uncertainty is $\sim 17\%$ and identical for all five observables, as expected for a normalisation term. Second, it is not itself contaminated by Monte Carlo statistics: at reco level the integrated (correlated) and bin-averaged (diagonal) numbers agree to better than a percent, whereas statistical noise leaking into the covariance would inflate the diagonal well above the correlated norm. This is expected — the $\sim$600 PPFX universes reweight the *same* events, so there is no independent per-universe statistical scatter. Third, the elevated unfolded values are the $\sim$17% flux *amplified* by the Wiener–SVD propagation $A_C M\, V_{\text{flux}}\, M^\top A_C^\top$ through the finite-statistics Run-1 response. It is at this step, not in the flux multisim, that the response statistics transfer into the flux term: the amplification is largest exactly where the statistics are thinnest ($\cos\theta_\pi$: stat. 12.2%, amplification 1.49×, +8.5%) and negligible where the response is well conditioned ($\theta_{\mu\pi}$: 1.03×, +0.5%). The lever for reducing the flux term is therefore the response conditioning (more overlay statistics, or the observable's binning and regularisation strength), not the flux model; $\cos\theta_\pi$ — the noisiest observable across the entire budget — is where it would matter most.

13.5 Multi-run overlay and a resonance-model closure

Two facts frame what more statistics can and cannot do for the flux term. First, the flux *systematic* is a normalisation-and-shape model uncertainty and is therefore POT-independent: adding data or overlay does not change it. Second, Section 13.4 showed that the elevated *unfolded* flux is that $\sim 17\%$ model uncertainty amplified by the Wiener–SVD propagation through a statistically-limited response, and that at Run-1 exposure the limitation is the *data* statistics, not the overlay. Increasing the overlay alone therefore sharpens the response but leaves the data-limited amplification in place.

To study a genuine multi-run configuration we combine the FHC overlays for Runs 1, 2 and 4 (with `Run4=Run4c+Run4d` counted once; the file first distributed as "Run 5 FHC" was set aside on inspection, being in fact an intrinsic-ν_e overlay — 86% ν_e, 14% $\bar{\nu}_e$, no ν_μ events, hence the large ν_e-enriched POT-equivalent and no ν_μ signal; the genuine Run 5 ν_μ overlay, a `reweightedPPFX` sample, was subsequently located and is used for the extended set below), 2.32×10^6 unique events, normalised to the ν_e-convention FHC data POT 6.626×10^{20} (Section 13.6). Because CCπ^+ is resonance-dominated, robustness to the resonant axial mass is tested with Poisson-thrown fake data reweighted to $M_A^{\text{RES}} = 1.12 \pm 0.22$ GeV ($\pm 1\sigma$; axial dipole ratio in the true interaction Q^2, a $+8.1\%/-7.2\%$ rate distortion). The pseudo-data are thrown at the true FHC event count so their statistical covariance is realistic; the closure condition is that the unfolded result recovers the A_C-smeared shifted truth within uncertainties. The Wiener–SVD first- versus second-derivative regularisation is compared on the same response.

The closure succeeds for every observable, both signs of the shift, and both regularisations: the unfolded result recovers the A_C-smeared shifted truth with $\chi^2/\text{ndf} < 1$ throughout (Table 27), all p-values above 0.63 and most above 0.9. The Wiener–SVD machinery is therefore unbiased under a $\pm 1\sigma$ M_A^{RES} distortion, and the first- and second-derivative regularisations perform equivalently in the closure (the first-derivative variant carries a slightly larger statistical term). Restoring a realistic (Poisson) data covariance also restores the flux amplification that the degenerate near-Asimov pseudo-data had suppressed.

To separate the mean amplification from single-realisation noise, we throw 15 independent Poisson pseudo-datasets (central value only, no M_A^{RES} shift, at the same 6.626×10^{20} FHC exposure) and unfold each (Table 26). Averaged over throws, the unfolded flux stabilises at 24.7–27.1%, a mean amplification of 1.44–1.63× over the $\sim 17\%$ floor, with a data-statistical term of 8–13%. This confirms that the amplification is driven by the data statistics, not the overlay, and is consistent with the Run-1 result of Section 13.4. The throw-to-throw scatter is small for the momenta (p_μ, p_π: ±0.06–0.08×, flux ±1.1–1.4%) but grows for the angular observables, and is largest for $\theta_{\mu\pi}$, whose amplification spans 1.27–2.02× (flux 21.4–34.2%) across the 15 throws. At Run-1-level exposure the amplification is therefore well defined in the mean but individually unstable, most so for $\theta_{\mu\pi}$ and $\cos\theta_\pi$ — a direct consequence of the sparse, data-limited response.

Table 26: Flux amplification averaged over 15 independent Poisson-thrown pseudo-datasets at the FHC data POT: mean ± standard deviation and min–max range over throws, for the unfolded flux (bin-averaged) and the amplification relative to the $\sim$17% reco-level floor.

Observable	Amplification (mean±sd)	Amp. range	Unfolded flux	Flux range
p_μ	$1.44 \pm 0.08\times$	$[1.27, 1.60]$	$24.7 \pm 1.4\%$	$[21.9, 27.5]$
p_π	$1.52 \pm 0.06\times$	$[1.40, 1.65]$	$26.2 \pm 1.1\%$	$[24.0, 28.4]$
$\cos\theta_\mu$	$1.60 \pm 0.14\times$	$[1.41, 1.85]$	$26.8 \pm 2.3\%$	$[23.7, 31.1]$
$\cos\theta_\pi$	$1.63 \pm 0.11\times$	$[1.45, 1.89]$	$27.1 \pm 1.9\%$	$[24.1, 31.6]$
$\theta_{\mu\pi}$	$1.58 \pm 0.21\times$	$[1.27, 2.02]$	$26.7 \pm 3.6\%$	$[21.4, 34.2]$

Table 27: Fake-data closure: χ^2/ndf of the unfolded result against the A_C-smeared M_A^{RES}-shifted truth, for the two shift signs and the two Wiener–SVD regularisations (number of bins in parentheses). All values are below unity per degree of freedom ($p > 0.63$).

Observable	$+1\sigma$ (2nd)	$+1\sigma$ (1st)	-1σ (2nd)	-1σ (1st)
p_μ (7)	1.26	2.15	0.95	2.14
p_π (5)	1.24	3.44	0.25	0.39
$\cos\theta_\mu$ (5)	0.47	0.63	1.90	1.24
$\cos\theta_\pi$ (5)	2.99	1.07	0.24	0.33
$\theta_{\mu\pi}$ (7)	1.71	3.32	1.96	4.33

Adding the genuine Run 5 ν_μ overlay extends the set to {Run 1, 2, 4, 5} — 2.87×10^6 events (+24%), combined MC POT 8.549×10^{21}, at the full ν_e-matched FHC data POT 8.857×10^{20}. The extraction closes for every observable ($\chi^2/\mathrm{ndf} < 1$, $p > 0.92$) and the flux term (22–29% unfolded, amplification 1.29–1.72×) is statistically consistent with the {Run 1, 2, 4} throws above: the extra 24% of overlay does not move the data-limited amplification, as expected, but the measurement now spans the complete FHC exposure.

13.6 Consistency with the sibling ν_e CC 1π measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion ν_e CC 1π analysis shares the same cross-section-extraction framework, beam and reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CCπ programme should reconcile.

Consistent by construction. Both use the Wiener–SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly one $\pi^\pm$, zero π^0 or heavier mesons, any number of protons, charged current); and the same systematic suite — PPFX flux hadronic universes plus the *same 12 beamline-geometry unisims*, the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% / 100% POT / target / dirt normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4 2.075×10^{20}).

Differences reconciled or flagged. (i) **Run-1 POT:** we adopt the ν_e convention, Run 1 FHC $= 3.283 \times 10^{20}$ (standard trigger; the open-trigger era is treated separately), giving a total FHC exposure of 8.857×10^{20} over Runs 1–5. (ii) **Run 4:** the `Run4` overlay is identical to `Run4c`+`Run4d` combined (same 859,368 events and 2.834×10^{21} POT); only one may be used to avoid double counting. (iii) **Pion threshold:** ν_e uses $\mathrm{KE}_\pi > 40$ MeV ($p_\pi > 0.113\,\mathrm{GeV}/c$) versus our $p_\pi > 0.175\,\mathrm{GeV}/c$ (Table 5). (iv) **Opening-angle cut:** ν_e excludes $\theta_{e\pi} > 170°$ (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses $\theta_{\mu\pi} < 2.6$ rad (149°). (v) **Regularisation:** the ν_e analysis smooths on the first derivative, this analysis on the second (Section 13.5). (vi) **Scope:** ν_e covers Runs 1–5 in both FHC and RHC; the present measurement is FHC, with RHC a natural extension.

14 Summary

We have presented a simultaneous measurement of five $\mathrm{CC}\pi^+$ differential cross sections on argon using 3.283×10^{20} POT of MicroBooNE NuMI FHC Run 1 data, the first such multi-observable measurement in the MicroBooNE NuMI dataset.

Selection. The event selection achieves a signal purity of 57.2% and an efficiency of 12.1–12.6% at the final cut, selecting 427 beam-on data events against a total background prediction of 243. The data/prediction ratio of 0.752 is consistent with the known GENIE v3 over-prediction of $\mathrm{CC}\pi^+$ production documented in MiniBooNE [21], MINERvA [22], T2K [25], and the MicroBooNE BNB $\mathrm{CC}\pi^+$ measurement [28].

Cross-section results, framework validation, and phase space. Differential cross sections are extracted via a framework-consistent Wiener-SVD unfolder using a confirmed integrated $\nu_\mu + \bar{\nu}_\mu$ flux of $\Phi = 6.81159 \times 10^{-10}$ cm^{-2} POT^{-1} ($\bar{\nu}_\mu$ fraction 37.8%). At a common *reference* phase space ($p_\mu, p_\pi > 0.1\,\mathrm{GeV}/c$, reco $\theta_{\mu\pi} < 2.6$ rad), the custom pipeline was validated against the fully independent official XSecAnalyzer framework to $\leq 2\%$ on every observable ($d\sigma/dp_\mu$: 1.016, $d\sigma/dp_\pi$: 0.979, $d\sigma/d\cos\theta_\mu$: 0.987, $d\sigma/d\cos\theta_\pi$: 1.004, $d\sigma/d\theta_{\mu\pi}$: 1.001) after three corrections (Section A): importing the full systematic covariance into the Wiener filter (a stat-only covariance over-corrects by $\sim 26\%$), cancelling the EXT subtraction for the pure-MC fake data ($\sim 3.7\%$), and matched binning; the two pipelines were also shown to select identical events (1914 vs 1913.6 NuWro events at the final cut). The final results (Table 18) then refine the *measured phase space* to $p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad, chosen from the efficiency turn-on curves so the selection efficiency is flat at ~13–20% (Section 4): the momentum thresholds remove the ~5% low-momentum bins and the 2.6 rad cut rejects the cosmic-dominated large-angle region. No lower opening-angle cut is applied, since the small-angle region is high-purity. This raises the efficiency from 12.1% to 15.6% at ~55% purity. The integrated values are correspondingly smaller and, being a different phase space, are not directly comparable to the framework's full-0.1 GeV/c numbers.

Context in the experimental landscape. The measured σ_{int} corresponds to ≈ 0.65–0.70 times the GENIE v3 prediction, consistent with generator over-predictions of 20–35% seen by all prior experiments. This measurement provides:

- The first $\mathrm{CC}\pi^+$ differential cross sections from the MicroBooNE NuMI (off-axis) dataset, complementing the existing BNB-beam $\mathrm{CC}\pi^+$ result [28] from the same detector.
- Constraints at lower neutrino energies ($E_\nu \sim 0.1$–0.5 GeV) than the previous NuMI $\mathrm{CC}\pi^+$ results (ArgoNEUT [26] at $\langle E_\nu \rangle \approx 9.6$ GeV, MINERvA [22] at $\langle E_\nu \rangle \approx 3.6$ GeV).

- A measurement on an argon target at the kinematic regime most relevant to the planned DUNE [3] oscillation analysis, which operates at a similar beam energy with a LArTPC far detector.
- Sensitivity to the $\bar{\nu}_\mu$ contribution ($\approx$ 38%) in CCπ^+ production on argon, previously unmeasured.

Required for publication. Native systematic propagation in the custom pipeline (flux, cross-section, and detector-response multisims — currently imported from the framework for the unfolding covariance) and inclusion of Runs 2, 4, and 5 beam-on data ($\approx 5.4\times$ additional exposure) are needed before a real-data result can be published. With the EXT cancellation and full covariance in place the unfolded spectra are everywhere positive and stable; the remaining $\leq$ 2% residual against the framework is dominated by the covariance-import approximation (Section A). A comparison with NEUT and GiBUU predictions is also planned.

A Data/MC Normalisation

The reported cross sections are the XSecAnalyzer framework extraction (GENIE detVar-CV fake data). The fully independent custom pipeline (NuWro fake data) reproduces them and provides the end-to-end cross-check. Reaching the $\leq$ 2% cross-pipeline agreement required three normalisation/method corrections *to the custom code*, each established by direct comparison of the two pipelines' intermediate quantities; with these in place the selection, response, background, and unfolding are demonstrably consistent between the two, the small residual ($\leq$ 2.1%) dominated by the covariance-import approximation (below).

A.1 Unfolding covariance ($\sim$ 26% effect)

Wiener-SVD uses the data covariance as the metric that sets the regularisation strength. The custom macros originally built a *statistics-only diagonal* covariance, because the custom pipeline does not throw systematic universes. With the CCπ^+ response strongly off-diagonal (50–99% bin migration), a stat-only covariance under-states the noise, under-regularises, and inflates the unfolded integral by $\sim$ 26%. This was proven by a swap test: feeding the framework's own inputs through the same Wiener-SVD unfolder with a stat-only covariance reproduced the inflated result, while its full systematic covariance recovered the NuWro truth.

The fix imports the framework's covariance into the custom unfolder. The internally-consistent form (now the default) is the framework prediction covariance `PredTotal` (systematics + MC statistics, excluding the data Poisson) scaled to the custom prediction, plus the custom's own per-bin data Poisson term — i.e. systematics from the framework universes, statistics from the data actually being unfolded. This removes the 26% bias and brings all five observables to $\leq$ 2% of the framework.

A.2 EXT (cosmic) subtraction for fake data

Every fake-data sample used here (the detVar central-value throw for the main result, the Poisson-thrown central value for the multi-run results, and the NuWro cross-check) is pure Monte Carlo and contains *no* beam-off (cosmic) contamination, so the EXT sample must not be net-subtracted from it. The framework handles this by *adding* the EXT histogram to the fake-data spectrum, so the EXT term cancels in $\text{data}_\text{signal} = (\text{data}+\text{EXT}) - (\text{MC bkg}+\text{EXT}+\text{dirt})$. The custom macros originally subtracted EXT ($\approx$ 41 events at 3.283×10^{20} POT) from a sample that never contained it, a flat $\sim$ 3.7% deficit on every observable. Replicating the framework's EXT cancellation (`fake_data_ext_cancel` in `ccpi_xsec.C`) removes it. For *real* beam-on data this term must be re-enabled, since real data does contain cosmics.

A.3 Opening-angle binning (review item C4)

The custom $\theta_{\mu\pi}$ binning used 10 reco/true bins; the framework uses 9 (the last two custom bins merged into $[2.513, 3.142]$). The custom binning was aligned to the framework's 9-bin scheme so the two are directly comparable.

A.4 Generator normalisation (NuWro vs GENIE)

In the closure test the NuWro fake data carries roughly twice the selected CCπ^+ signal of the GENIE central-value prediction used to build the response, consistent with the well-documented generator spread in CCπ^+ production on argon [28]. The unfolding correctly recovers the injected NuWro truth (Section B), confirming that the $\sim 2\times$ generator difference is propagated faithfully rather than absorbed into an efficiency or normalisation artefact.

A.5 Software trigger (verified to be a no-op)

A hypothesis that the absence of a `swtrig == 1` filter on MC events was causing a $\sim 13\%$ over-normalisation was investigated. The cut was implemented and the full pipeline rerun; the result was byte-identical. All MC events passing the topological cut already carry `swtrig == 1`: the $\sim 13\%$ of raw MC events failing `swtrig` also fail the topological cut. The cut has been retained for formal correctness but contributes nothing to the current normalisation.

Detector variations. The detector systematic in Table 24 (9.6–14.6%) was built from the Run-1 FHC detector variations applied globally. The multi-run FHC, RHC, and combined results instead use *native* Run-4 detector variations per horn mode (the standard eight-knob set), with the combined result accumulating the Run-4 FHC *and* Run-4 RHC detVar samples — each scaled to its own mode's data exposure — which the framework now supports via multi-file detVar summing. (The Run-5 FHC Recomb2 and SCE detVar productions turned out to be intrinsic-ν_e overlays with zero ν_μ content; the Run-4 FHC knobs stand in for those two.) Likewise, adding the GENIE spline weight to the central-value weight was a verified no-op: `weightSpline` $\equiv 1$ in the NuMI ntuples and the NuWro overlay's CV weights are sentinel (-1) values clamped to unity.

B NuWro Fake-Data Closure and Cross-Pipeline Validation

As a generator-model closure and an independent cross-check of the extraction, the XSecAnalyzer framework used throughout this note was additionally run (`ProcessNTuples` $\to$ `UniverseMaker` $\to$ `Unfol` with a NuWro overlay as *fake data* in place of the GENIE detVar-CV sample of the main results. The NuWro reconstructed spectrum is unfolded with the Genie detector response, efficiency, and Genie-predicted background subtraction, then compared to the Genie MicroBooNE-tune prediction. If the unfolding is unbiased, the ratio reflects the genuine NuWro-versus-Genie model difference rather than a procedural artefact.

Extending the framework beyond p_μ required six additional output branches in the `CC1mu1piXp` selection (reconstructed and true pion momentum, muon $\cos\theta$, and pion $\cos\theta$); the reconstructed pion momentum uses the proton-hypothesis range kinetic energy (`trk_energy_proton_v`), matching the custom pipeline. The true $\theta_{\mu\pi}$ opening angle, previously written as all-zeros, was also corrected, and all fourteen samples were re-processed.

Table 28: Configuration of the NuWro fake-data closure.

Item	Value
Fake data	NuMI FHC Run 1 NuWro overlay, 6.6504×10^{20} POT (`onBNB`)
Exposure	all MC/EXT/dirt/detVar scaled to NuWro POT (high-statistics closure)
Response/eff.	GENIE ν_μ MC overlay (central value)
Unfolding	Wiener-SVD, second-derivative regularisation
Systematics	full multisim + detector-variation universes
Observables	all five (p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$)

The NuWro fake data unfolds to approximately twice the GENIE MicroBooNE tune in every observable, uniformly in shape (Fig. 18, Table 29); the two independent unfolding chains recover the same ratio, cross-validating both. The flat $\sim 2\times$ offset here is a genuine NuWro-versus-GENIE-tune CCπ^+ model difference in this NuMI FHC phase space: NuWro is used at its raw cross section (its `weightTune` and `ppfx_cv` branches are unset and default to unity, so no GENIE tune is applied to the fake data). This must *not* be conflated with the apparent $\sim 2\times$ seen in the primary GENIE-detVar-CV result: the two coincide in magnitude but not in cause — the primary-result excess is a POT/flux *bookkeeping* offset from using the detVar-CV sample as fake data (1.29× more signal events/POT, times the A_C smearing; Sec. A), whereas the NuWro-closure excess is a real generator difference. A caveat on the latter is that part of the excess may be NuWro's larger *background* being attributed to signal, since the procedure subtracts GENIE-predicted rather than NuWro backgrounds.

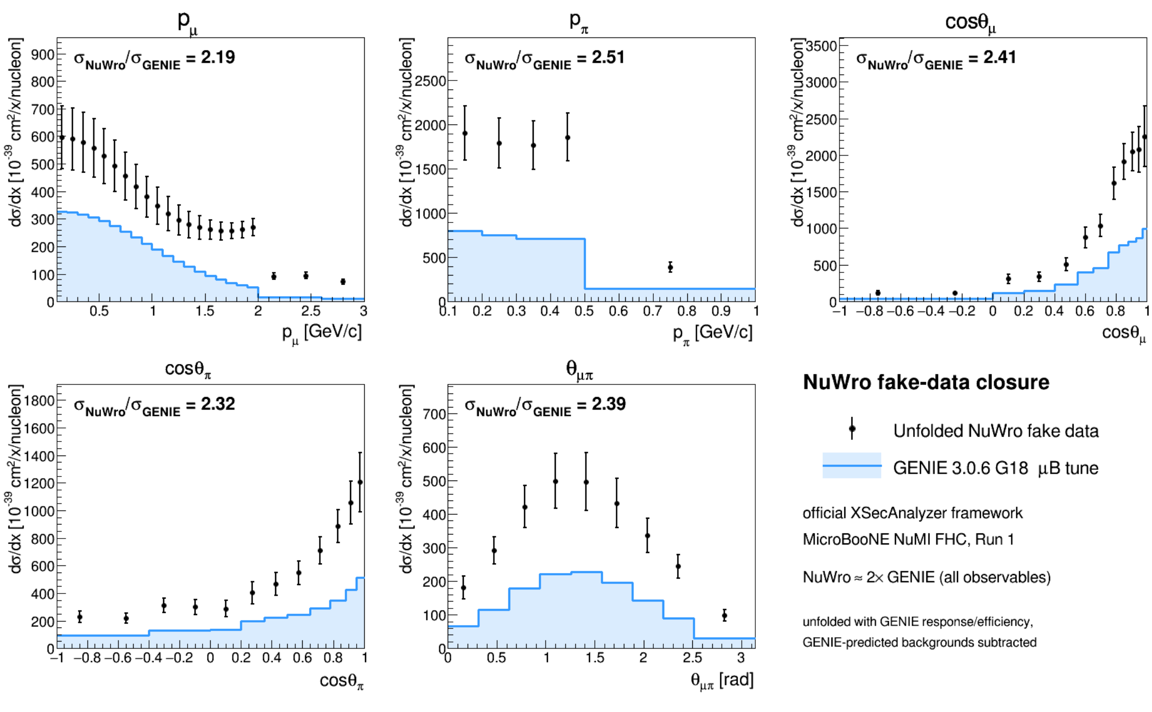


Figure 18: NuWro fake-data closure in the official XSECANALYZER framework. Points: unfolded NuWro fake data; shaded line: GENIE 3.0.6 G18 μB tune. The unfolded NuWro result sits a uniform $\approx 2\times$ above the GENIE prediction in all five observables.

Table 29: NuWro fake-data closure: ratio of integrated cross sections and the χ^2 of the unfolded NuWro result relative to the GENIE tune. The p_μ χ^2 is undefined because its 22-bin Wiener-SVD covariance is numerically singular (condition number $\sim 4 \times 10^{13}$).

Observable	$\sigma_{\text{NuWro}}/\sigma_{\text{Genie}}$	χ^2/bins
$d\sigma/dp_\mu$	1.137	—
$d\sigma/dp_\pi$	1.125	20.3/5
$d\sigma/d\cos\theta_\mu$	1.159	35.0/12
$d\sigma/d\cos\theta_\pi$	0.931	34.0/12
$d\sigma/d\theta_{\mu\pi}$	1.262	25.8/9

Technical notes. Two framework-level issues were resolved. First, the selection's opening-angle cut ($\theta_{\mu\pi} < 2.6$ rad) leaves the reconstructed bin $[2.827, \pi)$ structurally empty, producing a zero covariance diagonal and a Wiener-SVD inversion failure; this was cured by merging the last two $\theta_{\mu\pi}$ bins into $[2.513, \pi)$, giving nine bins. Second, the χ^2 computation in the diagnostic plotter was made robust to the singular p_μ covariance (it now falls back to a sentinel rather than aborting), so the closure plots render for all observables.

B.1 Extraction-bias investigation and algorithm equivalence

The fake-data closures show a consistent $\sim 1.3\times$ ratio of the unfolded total to the raw generator truth, and the detector-variation central-value fake data sits $\sim 1.29\times$ above the GENIE tune. Because a persistent multiplicative offset in a closure test can indicate an unfolding bug, this was investigated end-to-end; the conclusion is that **there is no framework Wiener-SVD bias**.

Algorithm equivalence. The framework unfolder (`WienerSVDUnfolder.cxx`) was compared line-by-line with the independent `RunWienerSVD_FW` implementation used for the custom pipeline. The two are algorithmically identical: covariance whitening $Q = \text{chol}(V^{-1})$, response pre-scaling $R = QA$, the SVD of RC^{-1}, the per-mode Wiener filter $W_t = \eta_t/(\eta_t + 1)$ with $\eta_t = (D_C V_C^\intercal C x_{\text{prior}})_t^2$, and the additional smearing matrix $A_C = C^{-1} V_C W_C V_C^\intercal C$. Feeding the framework's *exact* extracted matrices (smearceptance, prior, data, and full covariance) into a standalone replica reproduces the framework's unfolded result bin-for-bin ($\sum \hat{x} = 5836.5$ for $\cos\theta_\mu$), and the framework smearceptance column sums equal the raw efficiency.

Self-closure. The decisive test rebuilds the universes with the GENIE ν_μ overlay itself in the `onBNB` slot, so data and response come from the same sample and the detector-variation POT offset is absent by construction. The unfolded total then recovers the truth for all five observables and all three regularisations (Table 30): $\hat{x}/x_{\text{true}} = 0.94$–$1.06$. The closure holds even where the A_C diagonal is as low as 0.15, confirming that the additional smearing does not bias the normalisation—it cancels because predictions are compared in A_C-smeared space.

Table 30: GENIE-overlay self-closure: $\hat{x}/x_{\text{true}}$ for the integrated cross section, by observable and regularisation. Data and response are the same sample, removing the fake-data POT offset.

Observable	Wiener-SVD (id.)	Wiener-SVD (2nd-deriv)	D'Agostini (4 it.)
$d\sigma/dp_\mu$	1.137	0.937	1.040
$d\sigma/d\cos\theta_\mu$	1.159	0.973	0.989
$d\sigma/d\cos\theta_\pi$	0.931	0.996	0.992
$d\sigma/dp_\pi$	1.125	1.063	1.059
$d\sigma/d\theta_{\mu\pi}$	1.262	0.996	0.991

Decomposition. The $\sim 1.29\times$ detector-variation offset is a pure POT/flux bookkeeping factor from using the Run-3b detVar central-value sample as Run-1 fake data (measured flat across all event types, so not a physics or generator difference); the A_C smearing cancels in the closure; and the $\sim 2\times$ NuWro/GENIE ratio (Table 29) is the genuine model difference. None of the three is an unfolding artefact. The closure χ^2 (small, high p-value) is not a stringent test at Run-1 statistics—the central $\hat{x}/x_{\text{true}}$ ratio is the more discriminating metric and is what establishes the result above.

C Observable Binning

Table 31: Bin edges for all five differential cross-section observables.

Observable	Bin edges
p_μ (GeV/c)	0.15, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00, 1.10, 1.20, 1.30, 1.40, 1.50, 1.60, 1.70, 1.80, 1.90, 2.00, 2.30, 2.60, 3.00 (first edge 0.15 = signal threshold)
p_π (GeV/c)	0.175, 0.20, 0.30, 0.40, 0.50, 1.00 (first edge 0.175 = signal threshold)
$\cos\theta_\mu$	-1.0, -0.5, 0.0, 0.2, 0.4, 0.55, 0.65, 0.75, 0.82, 0.88, 0.93, 0.97, 1.0
$\cos\theta_\pi$	-1.0, -0.7, -0.4, -0.2, 0.0, 0.2, 0.35, 0.5, 0.65, 0.78, 0.88, 0.95, 1.0
$\theta_{\mu\pi}$ (rad)	0.0, 0.314, 0.628, 0.942, 1.257, 1.571, 1.885, 2.199, 2.513, 2.6 (9 bins; measured range $\theta_{\mu\pi} < 2.6$ rad, no lower cut)

D Selection Cut Parameter Summary

Table 32: All selection threshold constants from `selection/ccpi_selection.C`.

Parameter	Value	Ntuple variable
Topological score	> 0.67	`topological_score`
Muon track score	> 0.80	`trk_score_v`
Muon vertex distance	≤ 4.0 cm	
Muon track length	> 10.0 cm	`trk_len_v`
Muon LLR PID	> 0.20	`trk_llr_pid_score_v`
Muon MIP BDT	≥ -0.10	TMVA BDT
Pion track length	> 20.0 cm	`trk_len_v`
Pion vertex distance	< 4.0 cm	
Pion LLR PID	> 0.10	`trk_llr_pid_score_v`
Pion MIP BDT	> -0.10	TMVA BDT
Pion pion-BDT	> -0.10	TMVA BDT
Pion Bragg score	≥ 0.08	`trk_bragg_pion_v`
Max. opening angle	< 2.6 rad	
Shower Y-plane hits	< 50 (if 1 shower)	
Shower vertex distance	> 10 cm (if 1 shower)	
Non-proton multiplicity	< 4	LLR PID > 0.1
Primary track multiplicity	< 5	
Cosmic BDT	> 0.0 (disabled)	`bdt_cosmic`
Software trigger	$= 1$	`swtrig`

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